



TECHPIOASSET — ENTERPRISE ITAM REDESIGN

QA Test-Scenario Pack

Traceable test cases for every module — verified on Web AND Mobile

Functional · Negative · RBAC · Boundary · Concurrency · Security · Accessibility · E2E

Pack	QA Test-Scenario Pack (companion to the 86-page blueprint)
Audience	QA engineers, SDETs, test leads, release managers
Product	TechpioAsset — multi-tenant SaaS (Web + Mobile + REST API)
Golden rule	Every functional scenario is executed on BOTH the Web UI and the Mobile UI (and API where noted) before a task is called done.
Traceability	Scenarios derive from the redesign's state machines, lifecycles, validation rules, permission matrix, and notification/audit matrices.
Baseline	Monorepo at E:\TechpioAsset · live at https://piotask.com

HOW TO USE Each row is one executable test case (ID · Scenario · Type · Priority · Surface · Preconditions · Steps · Expected Result). Run P1 rows every release; P2/P3 on full regression. The **Surface** column tells you which client(s) to test — **Both** means run it on web and mobile.

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1. Test Strategy & Scope

1.1 Purpose & Objectives

This pack validates the redesigned TechpioAsset Enterprise ITAM platform's **authentication, authorization (RBAC/ABAC), and multi-tenant isolation** controls across all three delivery surfaces — Next.js 15 web, Expo/React Native mobile, and the NestJS v1/v2 REST API. The governing quality goal is **defense-in-depth verification**: every access decision is proven at the gateway/guard layer, the policy (ABAC) layer, and the PostgreSQL RLS backstop, not merely in the UI. Security controls are treated as functional acceptance criteria, not optional hardening.

1.2 Test Levels

Level	What it proves	Primary tooling target
Unit	Guard predicates, scope resolvers (<code>ALL/DEPARTMENT/DIRECT_REPORTS/OWN</code>), fail-closed default, no-self-escalation subset check, read-only mutation-deny middleware	Jest (NestJS providers), RN component logic
Integration	Guard → policy → RLS chain in one transaction; <code>SET app.tenant_id</code> GUC enforcement; SCIM mapping; token-family revoke	Nest test module + real Postgres w/ RLS
API / Contract	Endpoint-level authZ, RFC-9457 <code>problem+json</code> deny shape, idempotency, refresh rotation, step-up challenge headers	Supertest / Pact against <code>/api/v1</code>
E2E (UI)	Full user journeys: login → MFA → scoped dashboard → allowed/denied actions	Playwright (web), Detox/Maestro (mobile)
UI / Component	Conditional sidebar rendering by permission, disabled controls, biometric prompt, sync/conflict UX	Storybook + RTL / RN Testing Library

1.3 Environments

Env	Purpose	Notes
CI ephemeral	Unit + integration + contract on every PR	Seeded RLS-enabled Postgres; deterministic tenant fixtures
Staging	E2E web + mobile, SSO/SCIM against test IdP (Entra/Okta sandbox), smoke + contract	Blue-green mirror of prod config; real WAF/rate-limit
Security lane	Negative/IDOR/privilege-escalation, brute-force lockout, token-reuse	Runs isolated; no shared state with functional suites

1.4 Test Data & Tenants

- **Minimum three tenants:** `T-Alpha`, `T-Beta` (peer customers, must be mutually invisible), `T-MSP` (Super Admin fleet plane).
- **Full role coverage per tenant:** one seeded principal for each of the 13 canonical roles, plus one **custom role** and one `readonly:true` role for invariant tests.
- **Org tree** seeded so `DEPARTMENT`, `DIRECT_REPORTS`, and `OWN` scopes have non-trivial membership (a Dept Mgr with ≥2 reports and cross-department decoys).
- **Cross-tenant collision data:** identical asset tags / serials in `T-Alpha` and `T-Beta` to prove per-tenant unique keys and RLS row filtering.
- **Secrets/PII fixtures** are synthetic only; no real credentials. MFA seeds use fixed TOTP secrets for deterministic OTP generation.

1.5 Entry & Exit Criteria

Entry: build deployed to target env; RLS policies applied and migration-check green; test IdP reachable; seed fixtures loaded; all 13 role principals authenticate. **Exit:** 100% of P1 scenarios pass on **all applicable surfaces**; zero open S1/S2 defects; the Auditor read-only invariant, cross-tenant isolation, and no-privilege-escalation suites pass with **zero** exceptions (these are release-blocking regardless of severity math); traceability matrix shows every in-scope spec control mapped to ≥1 executed case.

1.6 Defect Severity & Priority

Severity	Definition
S1 Critical	Cross-tenant read/write, privilege escalation, auth bypass, Auditor mutation succeeds, refresh-reuse not revoked
S2 High	Scope leak within tenant (e.g., Employee sees others' assets), MFA/step-up bypassable, force-logout ineffective
S3 Medium	Correct backend denial but misleading UI (control shown-then-403), inconsistent error envelope
S4 Low	Cosmetic, copy, non-blocking UX on a secured path

Priority: **P1** = release-blocking security/happy-path (run every regression); **P2** = high-value negative/edge; **P3** = boundary/cosmetic. Any cross-tenant or escalation finding is S1 **and** P1 by definition.

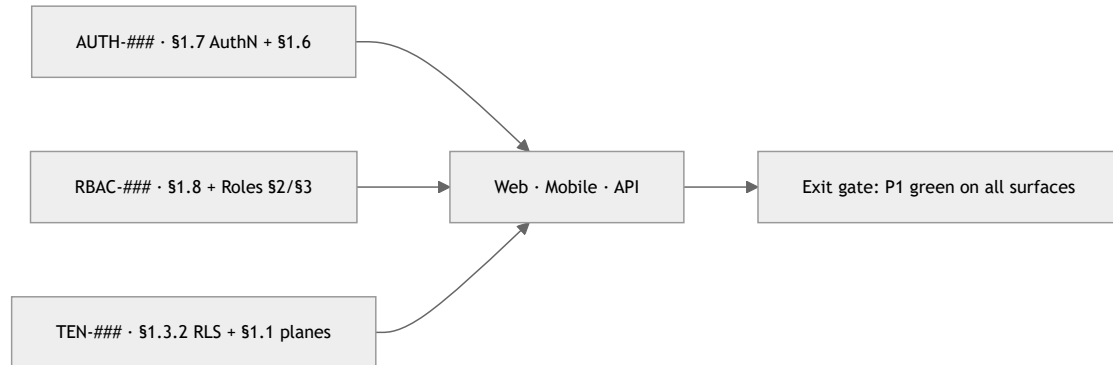
1.7 Dual-Surface Execution Rule (mandatory)

Every functional scenario is executed on **BOTH the Web UI and the Mobile UI, and against the API directly where an endpoint exists**. A pass on one surface never substitutes for another. Scenarios are tagged `Surface=Both`; where the mechanism genuinely differs (biometric unlock, offline-queued

mutation, PIN fallback, deep-link push), a **Mobile-specific twin** is written instead of collapsing it. UI-only passes are invalid for security controls — the authoritative assertion is always the API/gateway/RLS decision, with the UI verifying it surfaces correctly.

1.8 Traceability

Each case ID (**AUTH-###** , **RBAC-###** , **TEN-###**) maps to a spec clause: SRS §1.6 (zero-trust), §1.7 (auth), §1.8 (authZ), §1.3.2 (RLS), and Roles §1.3 (scopes), §2.12 (Auditor), §3.1/§3.3.1 (custom roles, invariants). Coverage map:



2. Authentication, RBAC & Multi-Tenant Isolation — Test Scenarios

2.1 Authentication (AUTH)

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
AUTH-001	Password login success (Argon2id) issues 15m access + 30d rotating refresh	Functional	P1	Both	Active user in T-Alpha , MFA off for this policy	1) Submit valid email+password 2) Inspect token response	200; access JWT TTL ~15m with sub/tenant_id/roles/perms/jti ; refresh issued, stored hashed per-device
AUTH-002	Invalid password is rejected without user enumeration	Negative	P1	Both	Active user	1) Submit valid email + wrong password 2) Submit unknown email	Both return identical generic 401; no timing/message difference revealing account existence
AUTH-003	Brute-force lockout after threshold	Security	P1	API	Lockout policy configured	1) Submit N failed logins for one account rapidly	Account/IP throttled per policy; 429 /lockout with Retry-After ; security audit event logged
AUTH-004	TOTP MFA challenge on password login	Functional	P1	Both	User with TOTP enrolled; tenant MFA=required	1) Pass password 2) Prompted for TOTP 3) Enter valid OTP	Access JWT issued only after valid OTP; wrong OTP → 401, no token
AUTH-005	WebAuthn/FIDO2 passkey MFA	Functional	P1	Web	Passkey registered for user	1) Password step 2) Complete WebAuthn assertion	Assertion verified server-side; JWT issued; replayed/forged assertion rejected
AUTH-006	MFA cannot be skipped by calling protected API with partial token	Negative	P1	API	MFA=required, pre-MFA interim token	1) Call a business endpoint with the pre-MFA token	401/403 problem+json ; interim token lacks full scope ; no data returned
AUTH-007	SSO via SAML 2.0 (SP-initiated)	Functional	P1	Both	Tenant IdP (SAML) configured	1) Choose SSO 2) Auth at IdP 3) Return with signed assertion	Assertion signature+audience validated; session established for correct tenant_id
AUTH-008	SSO via OIDC (IdP-initiated) with JIT provisioning	Functional	P1	Both	OIDC IdP; user not yet in tenant	1) Launch from IdP 2) First login	User JIT-provisioned, mapped to role per claims; session scoped to tenant
AUTH-009	SAML assertion for wrong tenant/audience rejected	Negative	P1	API	Assertion minted for T-Beta IdP	1) Replay assertion at T-Alpha ACS	Rejected; no session; audit records failed SSO with tenant mismatch
AUTH-010	SCIM 2.0 inbound provisioning creates user + role map	Functional	P1	API	SCIM token for T-Alpha	1) POST SCIM User with group → role mapping	User created in T-Alpha only, role assigned per group; idempotent on repeat
AUTH-011	SCIM deprovision disables access immediately	Security	P1	API	Active SSO user with live session	1) SCIM PATCH active=false (or DELETE)	User deactivated; existing sessions/refresh families revoked; next request 401
AUTH-012	Biometric login (Face/Touch) bound to secure-enclave device key	Functional	P1	Mobile	Device registered, biometric enrolled	1) Launch app 2) Face/Touch prompt 3) Succeed	Device-scoped key unlocks refresh; access JWT issued; key never leaves keystore
AUTH-013	PIN fallback when biometric fails/unavailable	Functional	P1	Mobile	Biometric enrolled, PIN set	1) Fail/cancel biometric 2) Enter valid PIN	Fallback grants session; wrong PIN after retries locks to full re-auth
AUTH-014	Biometric on an unregistered/rooted device denied	Negative	P1	Mobile	Device not in registry / integrity flag	1) Attempt biometric login	Denied; forced to full auth + device registration; no local token unlock
AUTH-015	Refresh-token rotation issues new refresh, invalidates old	Functional	P1	Both	Valid refresh token	1) Call refresh endpoint 2) Reuse the same old refresh	New access+refresh returned; the presented (now-rotated) token is invalid on reuse
AUTH-016	Refresh reuse detection revokes entire token family	Security	P1	Both	Rotated refresh token captured (stolen-token sim)	1) Rotate once 2) Replay the superseded refresh	Reuse detected → whole family revoked; both attacker and legit device must re-auth; security event logged
AUTH-017	Access token past 15m TTL is rejected	Boundary	P2	API	Access JWT aged beyond TTL	1) Call endpoint with expired JWT	401 problem+json ; must refresh; no grace acceptance
AUTH-018	Revoked jti in denylist blocks a still-unexpired token	Security	P1	API	Token logged out but not yet expired	1) Logout 2) Reuse the same access JWT	Denied via Redis jti denylist; 401
AUTH-019	Per-device session registry + admin force-logout	Security	P1	Both	User signed in on 2 devices	1) Admin force-logout device A 2) Device A acts	Device A sessions/refresh revoked and 401; device B unaffected
AUTH-020	Concurrent-session limit enforced	Boundary	P2	Both	Policy caps N concurrent sessions	1) Exceed the cap with N+1 logins	Oldest session evicted or new login refused per policy; registry consistent
AUTH-021	Step-up MFA required for sensitive op (asset disposal)	Security	P1	Both	Authenticated user with assets:asset:dispose , no recent step-up	1) Invoke dispose 2) Prompted for step-up 3) Provide factor	Op blocked until step-up satisfied; success only after fresh factor; audit captures step-up
AUTH-022	Step-up MFA for bulk delete / cost export	Security	P1	Both	User with export/bulk perms	1) Trigger cost export or bulk delete	Server demands step-up challenge; unfulfilled challenge → 403; no export produced
AUTH-023	Mobile step-up biometric for disposal	Security	P1	Mobile	IT Tech with dispose right	1) On device, tap Dispose	Fresh biometric prompt (step-up) required even within active session; cancel = op aborted
AUTH-024	Offline-queued sensitive op still enforces step-up on sync	Security	P1	Mobile	Dispose queued while offline	1) Queue dispose offline 2) Reconnect → replay	Server rejects replayed dispose lacking valid step-up context; op flagged for re-auth, not silently applied

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
AUTH-025	Idempotent refresh/login replay does not mint duplicate sessions	Boundary	P2	API	Idempotency-Key on POST	1) Replay identical login/refresh POST within 24h	Same first result returned; no second refresh family created

2.2 RBAC — Roles, Scopes & Invariants (RBAC)

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
RBAC-001	Deny-by-default: permission absent = 403	RBAC	P1	Both	Employee (scope OWN)	1) Call procurement:po:approve	403 problem+json ; no state change; fail-closed with no matching grant
RBAC-002	Missing scope resolves to OWN (fail-closed), not ALL	RBAC	P1	API	Role granted a read string with no scope attached	1) Read a list endpoint	Only the caller's own rows returned; never tenant-wide
RBAC-003	Employee OWN scope — sees only own assets	RBAC	P1	Both	Employee with 2 assigned assets; decoys owned by peers	1) Open My Assets / GET assets	Exactly the 2 owned rows; peers' assets absent on list, detail, and search
RBAC-004	Employee cannot read a peer's asset by direct ID (IDOR)	Negative	P1	Both	Peer asset ID known	1) Navigate/GET /assets/{peerId}	403/404; no record body; attempt audited
RBAC-005	Department Manager DEPARTMENT view + DIRECT_REPORTS approval	RBAC	P1	Both	Dept Mgr with 2 reports; other-dept decoys	1) View dept assets 2) Approve a report's request 3) Attempt approve for non-report	Sees dept rows; approves only reports; approval on non-report → 403
RBAC-006	Office Admin site (DEPARTMENT /SITE) scope confinement	RBAC	P2	Both	Office Admin over Site-1	1) Check-out at Site-1 2) Attempt Site-2 action	Site-1 succeeds; Site-2 asset invisible/denied
RBAC-007	Office Admin approval limited to low-value/site-local requests	Boundary	P2	Both	Threshold configured	1) Approve below-threshold local request 2) Approve above-threshold	Below passes; above → 403 requiring higher role
RBAC-008	IT Technician write is assignment-scoped, read is tenant-wide	RBAC	P1	Both	IT Tech; asset on-queue + off-queue	1) Read any asset 2) Update on-queue asset 3) Update off-queue asset	Reads succeed tenant-wide; on-queue update OK; off-queue update 403
RBAC-009	IT Manager full IT modules, blocked on finance/HR mutate	RBAC	P2	Both	IT Manager	1) Manage license/discovery 2) Attempt invoices:approve / people:manage	IT ops succeed; finance/HR mutations 403
RBAC-010	Finance Manager approves invoices, read-only on assets/procurement	RBAC	P2	Both	Finance Manager	1) Approve invoice (3-way match) 2) Attempt assets:asset:update	Invoice approve OK; asset write 403; asset financial fields readable
RBAC-011	Procurement Manager PO lifecycle + vendor-portal invite	RBAC	P2	Both	Procurement Manager	1) Raise/dispatch PO, invite vendor 2) Attempt finance:depreciation:run	Procurement + invite OK; finance run 403
RBAC-012	Inventory Manager stock/GRN full; cannot approve requests	RBAC	P2	Both	Inventory Manager	1) Receive GRN, adjust stock 2) Attempt requests:approve	Inventory ops OK; approval 403
RBAC-013	HR Manager people-scope; no financial cost visibility	RBAC	P2	Both	HR Manager	1) Trigger onboarding, read custody 2) Attempt read of asset purchase cost	People ops OK; cost fields serialized-out/denied
RBAC-014	Super Admin cannot read tenant business data without impersonation	Security	P1	API	Super Admin, no active impersonation session	1) GET T-Alpha asset directly	Denied; only a started, audited impersonation session grants gated read
RBAC-015	Super Admin impersonation is audited start→stop	Security	P1	Both	Super Admin, MSP membership over T-Alpha	1) Start impersonation 2) Read data 3) Stop	Data readable during session; impersonation start/stop + actions written to security audit
RBAC-016	Super Admin barred from tenants outside contract (MSP membership gate)	Negative	P1	API	Super Admin without membership over T-Beta	1) Attempt impersonate/read T-Beta	Denied even with ALL ; membership check fails closed
RBAC-017	Auditor read-only invariant — every mutation blocked at gateway	Security	P1	Both	Auditor (readonly:true), scope ALL read	1) Attempt create/update/delete/approve/execute across modules	All rejected by deny-all-mutations middleware before permission resolution; 403; zero writes
RBAC-018	Auditor invariant cannot be overridden by adding permission strings	Security	P1	API	Auditor role with a mutate string artificially attached	1) Attempt the granted mutation	Still denied — invariant precedes string resolution; regression-locked by standing test
RBAC-019	Auditor scan-to-verify writes only to append-only findings store	Functional	P1	Mobile	Auditor on device	1) Scan asset to verify	Verification event lands in immutable findings log; asset record and history untouched
RBAC-020	Auditor evidence export succeeds (signed)	Functional	P2	Both	Auditor	1) Run audit:evidence:export	Signed CSV/JSON produced; export is read/append-only, no operational mutation
RBAC-021	Custom role composes catalog strings + per-group scope	Functional	P2	Both	Company Admin	1) Build custom role from catalog with DEPARTMENT scope 2) Assign user	Role evaluates through same engine; user gets exactly composed perms at chosen scope
RBAC-022	No self-escalation — role cannot grant perms granter lacks	Security	P1	Both	Company Admin editing a role	1) Attempt to grant platform:* or a perm admin lacks	Rejected; grantable set is a subset of granter's effective perms; audited

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
RBAC-023	platform:* never grantable to any tenant/custom role	Negative	P1	API	Company Admin	1) Add platform:tenant:provision to a custom role	Rejected by plane-isolation guard; role save fails validation
RBAC-024	SoD conflict warning on toxic permission pair	Functional	P2	Web	Company Admin building role	1) Add procurement:po:approve + invoices:approve to one role	SoD conflict warning surfaced before save; requires explicit acknowledgment
RBAC-025	Vertical escalation blocked — Employee cannot self-assign admin role	Negative	P1	Both	Employee	1) Attempt admin:role:manage / self role change	403; role assignment endpoint denies; audited
RBAC-026	ABAC location condition — assign only within managed locations	RBAC	P2	Both	Site Manager with managedLocations={L1}	1) Assign asset at L1 2) Assign asset at L2	L1 assign OK; L2 assign denied by ABAC policy layer despite base permission
RBAC-027	Sidebar/nav renders conditionally by permission (no dead controls)	Functional	P3	Both	Employee vs IT Manager	1) Compare rendered nav	Only permitted modules shown; hidden items also 403 at API (UI hide ≠ sole control)
RBAC-028	Vendor confined to walled portal, zero tenant-app access	RBAC	P1	Both	Vendor principal, vendor_id bound	1) Attempt to reach any tenant app route/API	Denied; only vendor-portal:* on own vendor_id records; no CMDB/asset visibility
RBAC-029	Vendor sees only records linked to own vendor_id	Negative	P1	Both	Two vendors, overlapping tenant	1) Vendor A GETs Vendor B's PO/RMA	403/empty; strict vendor_id filter; no leakage of another supplier's records
RBAC-030	API-key scope constrains integration token (OAuth-style)	RBAC	P2	API	Key scoped read:assets	1) Call write inventory with the key	403; scope excludes writes; interactive JWT unaffected

2.3 Multi-Tenant Isolation (TEN)

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
TEN-001	No cross-tenant read — T-Alpha user cannot read T-Beta asset by ID	Security	P1	Both	Known T-Beta asset ID; T-Alpha Company Admin	1) GET /assets/{betaId} as T-Alpha	404/403; zero rows; even Company Admin ALL is tenant-bounded
TEN-002	No cross-tenant write — mutation targeting foreign tenant row fails	Security	P1	API	T-Alpha token, T-Beta record id	1) PATCH/DELETE T-Beta record	Denied; no write; audit records cross-tenant attempt
TEN-003	RLS backstop — forged/absent tenant_id context returns no rows	Security	P1	API	Query path with tampered/omitted app.tenant_id GUC	1) Force a query without correct SET app.tenant_id	RLS policy yields empty set; "even a buggy query cannot cross tenants" holds
TEN-004	Tenant claim in JWT cannot be swapped to reach another tenant	Security	P1	API	Valid T-Alpha JWT, edited tenant_id claim	1) Present tampered JWT	Signature invalid → 401; even if re-signed by test, gateway tenant-resolve + RLS block access
TEN-005	Per-tenant unique keys — duplicate asset tag allowed across tenants	Functional	P1	API	Same asset tag in T-Alpha and T-Beta	1) Create tag AST-100 in both tenants	Both succeed; uniqueness is (tenant_id, tag) , not global; no collision error
TEN-006	Per-tenant unique key enforced within a tenant	Negative	P2	Both	AST-100 exists in T-Alpha	1) Create second AST-100 in T-Alpha	Rejected with uniqueness error scoped to tenant
TEN-007	List/search never bleeds foreign-tenant rows	Security	P1	Both	Colliding data in both tenants	1) Run asset list, global search, saved filter as T-Alpha	Only T-Alpha rows; counts, aggregates, and dashboard tiles exclude T-Beta
TEN-008	Redis cache keyed per tenant — no cross-tenant cache hit	Security	P1	API	Same logical key in both tenants	1) Warm T-Alpha cache 2) Request as T-Beta	t:{tenantId}: prefix isolates; T-Beta gets its own value/miss, never T-Alpha 's
TEN-009	Object-storage / export scoped to tenant	Security	P1	Both	T-Alpha export job produced a file	1) T-Beta user requests that export URL/id	Denied; storage path tenant-scoped; no cross-tenant download
TEN-010	Webhook/domain event carries correct tenant_id , not delivered cross-tenant	Security	P2	API	Subscriptions in both tenants	1) Trigger event in T-Alpha	Only T-Alpha subscription receives it; payload tenant_id =Alpha; HMAC valid for Alpha secret
TEN-011	Offline mobile sync cannot apply an op to a foreign tenant	Security	P1	Mobile	T-Alpha device with queued op referencing T-Beta entity	1) Reconnect → /sync/batch replay	Op rejected; tenant context of device enforced server-side; no cross-tenant apply
TEN-012	Delta pull /sync/changes returns only device-tenant scoped records	Security	P1	Mobile	T-Alpha device	1) Pull changes since cursor	Only T-Alpha (and in-scope) rows; no T-Beta records in the delta
TEN-013	Rate-limit buckets isolated per tenant (no noisy-neighbor bypass)	Boundary	P2	API	T-Beta saturating its bucket	1) Flood as T-Beta 2) Normal calls as T-Alpha	T-Alpha unthrottled; buckets are (tenant, key, route-class)
TEN-014	Super-Auditor cross-tenant read confined to MSP-contracted tenants only	Security	P1	API	Super-Auditor over {T-Alpha} only	1) Read T-Alpha (allowed) 2) Read T-Beta (not contracted)	Alpha read OK (read-only); Beta denied by membership gate; still zero mutation anywhere

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
TEN-015	Cross-tenant IDOR via mobile deep-link push is neutralized	Negative	P1	Mobile	Deep-link crafted to a T-Beta entity delivered to T-Alpha device	1) Open deep link	App/API resolve tenant from session, not link; foreign entity → 403/not-found, no render

Coverage note: 70 scenarios across the three ID groups (25 AUTH, 30 RBAC, 15 TEN), weighted to P1 happy-path plus high-value negatives. Dual-surface is satisfied via Surface=Both where flows are identical and dedicated Mobile twins (AUTH-012/013/014/023/024, RBAC-019, TEN-011/012/015) where the mobile mechanism — biometric, PIN, offline replay, deep-link — diverges from web.

3. Asset Lifecycle & Status-Model — Test Scenarios

Legend — Type ∈ Functional | Negative | RBAC | Boundary | E2E | Data-integrity · Pri ∈ P1|P2|P3 · Surface ∈ Web | Mobile | API | Both. Status dimensions: LifecycleState (PLANNED/IN_PROCUREMENT/IN_STOCK/DEPLOYED/IN_MAINTENANCE/RETIRED/DISPOSED), AvailabilityState (AVAILABLE/RESERVED/ASSIGNED/IN_TRANSIT/IN_REPAIR/LOST), ConditionGrade (NEW/GOOD/FAIR/POOR/DAMAGED/END_OF_LIFE), OwnershipType (OWNED/LEASED/RENTED/BYOD/LOANER).

3.1 Create / Register & Identity (asset-tag, serial, QR)

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
AST-001	Manual asset registration seeds all four dimensions + history	Functional	P1	Both	Tenant T1; user has asset:create ; ProductModel & Category exist	1. Open New Asset 2. Set model, category, ownership=OWNED, assetTag=LAP-001 3. Save	Asset created with lifecycle=IN_STOCK, availability=AVAILABLE, condition=NEW (defaults), ownership=OWNED; version=0; one AssetLifecycleEvent (toState=IN_STOCK, reason set) and one initial ConditionLog (grade=NEW) written
AST-002	Create asset from received stock (GRN → asset instance)	E2E	P1	Web	GRN posted against a PO line for qty 3, serials captured	1. Open GRN receiving 2. Provision assets from GRN line 3. Confirm	3 Asset rows spawned, each linked to purchaseOrderLineId , lifecycle=IN_STOCK, availability=AVAILABLE, condition=NEW, ownership=OWNED, costBasis carried from PO line; lifecycle event reason references GRN
AST-003	Duplicate asset-tag within same tenant rejected	Negative	P1	Both	Asset LAP-001 already exists in T1	1. Create new asset 2. Set assetTag=LAP-001 3. Save	409/validation error citing UNIQUE (tenantId, assetTag) ; no asset, no lifecycle event created
AST-004	Same asset-tag reused across two tenants is allowed	Data-integrity	P2	API	LAP-001 exists in T1; caller authenticated in T2	1. POST asset assetTag=LAP-001 as T2	201 Created in T2; tenants isolated; no collision with T1's LAP-001
AST-005	Duplicate serial within tenant rejected; two active assets cannot share a serial	Negative	P1	Both	Asset with serial SN-9 exists (active, deletedAt NULL) in T1	1. Create asset serial=SN-9 2. Save	Rejected by partial UNIQUE (tenantId, serialNumber) WHERE serialNumber IS NOT NULL (BR-20); error cites serial conflict
AST-006	Serial optional — asset without serial (BYOD/legacy) saves	Boundary	P2	Both	ownership=BYOD selected	1. Create asset, leave serial empty 2. Save	Asset created with serialNumber NULL; partial unique index permits multiple NULL serials in same tenant
AST-007	Asset-tag is required	Negative	P2	Both	New asset form open	1. Leave assetTag blank 2. Save	Validation error "asset tag required"; no row created
AST-008	Serial uniqueness ignores soft-deleted asset	Boundary	P3	API	Asset SN-9 soft-deleted (deletedAt set)	1. Create new asset serial=SN-9	Allowed — partial unique still applies but is scoped by extension default filter; verify no active duplicate exists and creation succeeds (or documented conflict if kept). Pass = behavior matches spec: only one active holder of a serial
AST-009	QR/asset-tag label generation produces unique per-tenant payload	Functional	P2	Web	Asset registered, untagged	1. Open asset 2. Generate QR/label 3. Print	Printable QR + barcode generated; QR payload unique within tenant (BR-20); asset marked Tagged; label renders assetTag + serial
AST-010	Mobile scan-to-open resolves the correct asset	Functional	P1	Mobile	Asset LAP-001 tagged; scanner permission granted	1. Open mobile scanner 2. Scan LAP-001 QR	App opens LAP-001 detail with live four-dimension status chips
AST-011	Scanning a QR from a different tenant is not resolvable	Negative	P1	Mobile	Logged into T2; QR belongs to T1 asset	1. Scan T1 QR	"Asset not found in this tenant" — no cross-tenant leak (BR-03); no detail shown
AST-012	Scan of malformed / non-asset QR handled gracefully	Negative	P3	Mobile	Random QR (URL)	1. Scan unrelated QR	Friendly "unrecognized code" message; no crash; no navigation
AST-013	List & detail views render four independent status chips on both surfaces	Functional	P2	Both	≥ 1 asset exists	1. Open asset list 2. Open a detail	Each row/detail shows lifecycle, availability, condition, ownership as four separate labels (not one merged status)

3.2 Four Independent Status Dimensions & Guardrails

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
AST-014	Dimensions are orthogonal — condition change does not touch lifecycle	Functional	P1	Both	Asset DEPLOYED/ASSIGNED/GOOD/LEASED	1. Log condition=POOR	conditionGrade=POOR; lifecycleState, availabilityState, ownershipType unchanged; new ConditionLog row appended
AST-015	availability=ASSIGNED requires an active AssetAssignment	Negative	P1	API	Asset IN_STOCK/AVAILABLE, no active assignment	1. PATCH availabilityState=ASSIGNED directly	Rejected by guardrail §1.4 — "ASSIGNED requires active assignment + assignedToUserId"; state unchanged
AST-016	Setting availability=ASSIGNED succeeds when assignment exists	Functional	P1	API	Active AssetAssignment row present, assignedToUserId set	1. Assign flow completes	availabilityState=ASSIGNED accepted; assignedToUserId populated; guardrail satisfied

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
AST-017	RETIRED asset blocks new assignment	Negative	P1	Both	Asset lifecycle=RETIRED	1. Attempt Assign to user U	Blocked (\$1.4 / BR-21): "cannot assign a RETIRED/DISPOSED asset"; no assignment row; availability unchanged
AST-018	DISPOSED asset blocks new assignment and any transition back to active	Negative	P1	Both	Asset lifecycle=DISPOSED	1. Attempt Assign 2. Attempt set lifecycle=DEPLOYED	Both rejected (illegal transition Disposed→In-Use, BR-21); errors cite guarded state machine
AST-019	RETIRED/DISPOSED must be unassigned or availability=LOST	Data-integrity	P2	API	Attempt to retire an asset still availability=ASSIGNED	1. PATCH lifecycle=RETIRED while ASSIGNED	Rejected unless asset unassigned first or availability=LOST (\$1.4); guardrail enforced
AST-020	END_OF_LIFE condition surfaces soft retire recommendation (not hard block)	Functional	P2	Both	Asset condition set to END_OF_LIFE, lifecycle=IN_STOCK	1. Save condition=END_OF_LIFE	Save succeeds; UI surfaces non-blocking "recommend transition to RETIRED" advisory (\$1.4 soft rule); no forced state change
AST-021	IN_MAINTENANCE + still-ASSIGNED is warned, not blocked	Boundary	P2	Web	Asset DEPLOYED/ASSIGNED, flagged for maintenance	1. Set lifecycle=IN_MAINTENANCE while keeping assignment	Allowed with warning; typical pairing IN_MAINTENANCE→IN_REPAIR suggested but not enforced (\$1.4)
AST-022	Location XOR with-user enforced (cannot be both)	Negative	P1	API	Asset with currentLocationId set	1. PATCH assignedToUserId while currentLocationId still set	Rejected by CHECK NOT (currentLocationId IS NOT NULL AND assignedToUserId IS NOT NULL) ; one placement only
AST-023	Assigning to user clears currentLocationId (placement flips to with-user)	Functional	P1	Both	Asset AVAILABLE at Warehouse-BLR bin	1. Assign to user U 2. Accept	assignedToUserId=U, currentLocationId=NULL; placement is "with-user"; XOR invariant holds
AST-024	Return to stock clears assignedToUserId and sets a location	Functional	P1	Both	Asset ASSIGNED to U	1. Return to Warehouse-BLR bin	assignedToUserId=NULL, currentLocationId set; XOR invariant holds
AST-025	Cross-tenant location reference structurally impossible	Data-integrity	P1	API	Location L belongs to T2; caller in T1	1. PATCH T1 asset currentLocationId=L	Rejected — composite FK references Location(tenantId,id); T1 asset cannot point at T2 location (\$4.1)
AST-026	Invalid enum value rejected	Negative	P3	API	—	1. PATCH lifecycleState="ACTIVE" (not in enum)	400 validation error; only closed-enum values accepted
AST-027	Ownership orthogonal to availability — LOANER can be AVAILABLE	Boundary	P3	API	Loaner pool asset returned	1. Set ownership=LOANER, availability=AVAILABLE	Accepted; no guardrail couples ownership to availability

3.3 Assign → Acknowledge → Operate → Transfer → Return → Maintain → Retire → Dispose (E2E)

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
AST-028	Full happy-path lifecycle from stock to disposal	E2E	P1	Both	Asset IN_STOCK/AVAILABLE/NEW/OWNED at Warehouse-BLR	1. Assign to U 2. U e-signs 3. Operate 4. Transfer to V 5. V e-signs 6. Return 7. Maintenance 8. Retire 9. Sanitize 10. Dispose	Each step advances the right dimension(s); lifecycle NEW→IN_STOCK→DEPLOYED→...→RETIRED→DISPOSED; every transition writes an AssetLifecycleEvent; final state lifecycle=DISPOSED, disposal certificate attached (BR-18)
AST-029	Assignment not complete until assignee e-signs acceptance	Functional	P1	Both	Asset assigned to U, acceptance timer running	1. Assign 2. Do NOT sign	availabilityState=ASSIGNED but assignment.status=pending-acceptance; asset not yet "Active/In-Use" (BR-15); acknowledgedAt NULL
AST-030	E-sign acceptance binds custody and activates	Functional	P1	Mobile	Pending assignment for U; U on mobile	1. Scan asset 2. Review terms 3. E-sign	ESignature row created, assignment.acknowledgedAt set, eSignatureId linked; lifecycle=DEPLOYED, availability=ASSIGNED ("Active/In-Use")
AST-031	Unaccepted assignment expires and reverts to stock	Negative	P1	API	Assignment pending past configured acceptance window	1. Advance clock past window 2. Run expiry job	Assignment expired; asset reverts availability=AVAILABLE, lifecycle=IN_STOCK, assignedToUserId cleared (BR-15); lifecycle event records revert
AST-032	Assignee cannot e-sign for a different user's assignment	RBAC	P1	Mobile	Assignment is for U; W is authenticated	1. W attempts to e-sign U's assignment	Rejected — acceptance must be by the assignee (BR-02 scope); no signature recorded
AST-033	Transfer user→user requires re-acceptance by new custodian	Functional	P1	Both	Asset ASSIGNED to U, accepted	1. Transfer to V	New AssetTransfer (fromUserId=U,toUserId=V); asset enters pending re-acceptance; not "Active" for V until V e-signs (BR-15)
AST-034	Transfer site→site sets IN_TRANSIT then received	Functional	P1	Both	Asset at Site-A	1. Initiate transfer to Site-B with carrier+tracking 2. Ship 3. Receive at Site-B	availabilityState=IN_TRANSIT after ship (shippedAt set); on receive currentLocationId=Site-B, receivedAt set, availability returns to prior/AVAILABLE
AST-035	In-transit asset cannot be assigned to a user	Negative	P2	API	Asset availability=IN_TRANSIT	1. Attempt Assign to user	Rejected — asset in transit is not handable; availability unchanged

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
AST-036	Return captures condition-on-return	Functional	P1	Both	Asset ASSIGNED to U, condition=GOOD	1. Return, set conditionOnReturn=FAIR 2. Confirm receipt	AssetReturn row (conditionOnReturn=FAIR, returnedById, receivedById); new ConditionLog grade=FAIR appended; Asset.conditionGrade cache updated to FAIR; asset back IN_STOCK/AVAILABLE
AST-037	Damaged-on-return routes to maintenance, not available pool	Functional	P2	Both	Asset returned with conditionOnReturn=DAMAGED	1. Return with DAMAGED	Asset lifecycle=IN_MAINTENANCE / availability=IN_REPAIR (not AVAILABLE); ConditionLog=DAMAGED; advisory to open MaintenanceTicket
AST-038	Maintenance handoff → repair → back to active/stock	E2E	P2	Both	Asset condition=POOR, availability=ASSIGNED	1. Open MaintenanceTicket 2. Set IN_MAINTENANCE/IN_REPAIR 3. Complete repair, condition=GOOD	On close: ConditionLog=GOOD, availability returns (ASSIGNED or AVAILABLE), lifecycle back to DEPLOYED/IN_STOCK; lifecycle events bracket the maintenance window
AST-039	Retire requires approval and legal transition	Functional	P2	Web	Asset EoL, unassigned; user has asset:retire	1. Retire with reason + approval	lifecycle=RETIRED, condition=END_OF_LIFE, removed from active pool; retiredAt set; lifecycle event with actor+reason
AST-040	Dispose blocked without sanitisation + disposal certificate	Negative	P1	Both	Asset RETIRED, no certificate	1. Attempt Dispose	Blocked (BR-18): "data-sanitisation record + disposal certificate required"; state stays RETIRED
AST-041	Dispose succeeds with certificate attached	Functional	P1	Both	Asset RETIRED, sanitisation cert + disposal cert uploaded	1. Dispose with certificates	lifecycle=DISPOSED; certificates linked as AssetDocuments; final immutable lifecycle event; asset no longer assignable

3.4 Append-only History, Optimistic Locking, Custody Chain

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
AST-042	Lifecycle history is append-only and ordered	Data-integrity	P1	API	Asset with ≥3 transitions	1. GET lifecycle timeline	Every transition present as immutable AssetLifecycleEvent (fromState, toState, actor, reason, occurredAt) ordered DESC; no prior row mutated
AST-043	Attempt to UPDATE/DELETE a lifecycle event is rejected	Negative	P1	API	Existing AssetLifecycleEvent row	1. Attempt PATCH/DELETE the event	Rejected by REVOKE UPDATE/DELETE + trigger (\$4.2); ledger unchanged; corrections require a new compensating row
AST-044	Condition history reconstructs correctly; Asset.conditionGrade = latest log	Data-integrity	P1	Both	Asset with condition NEW→GOOD→POOR logged	1. GET condition history 2. Read asset detail	ConditionLog has 3 append-only rows in order; Asset.conditionGrade cache = POOR (latest); no back-dating alters earlier rows
AST-045	Optimistic-locking concurrent edit yields 409	Data-integrity	P1	API	Two clients load same asset version=5	1. Client A PATCH (version 5→6) succeeds 2. Client B PATCH with stale version=5	A: 200, version=6. B: 409 Conflict (update count=0); B's write discarded (\$4.3); no lost update
AST-046	Concurrent assign attempts on one asset — only one wins	Data-integrity	P1	Both	Asset AVAILABLE; two managers assign simultaneously	1. Both submit Assign at once	Exactly one assignment succeeds; the other gets 409/"already assigned"; asset ends with a single active assignment; no double-custody
AST-047	Stale-version conflict surfaced to user with reload prompt	Negative	P2	Web	Asset edited elsewhere after page load	1. Edit condition on stale detail 2. Save	UI shows "record changed, please reload" from 409; edits not silently applied
AST-048	Custody chain is complete and gap-free	Data-integrity	P1	Both	Asset assigned U→transferred V→returned	1. View custody chain	Ordered chain: assignment(U, acknowledged, eSign) → transfer(U→V) → acceptance(V, eSign) → return; each link timestamped with actor + e-signature where required (BR-16); chain reconstructable for audit (J4)
AST-049	Custody chain preserved after GDPR pseudonymisation	Data-integrity	P2	API	Leaver U had custody; erasure request processed	1. Pseudonymise U's PII 2. View chain	Chain intact and immutable; U's identity pseudonymised but assignment/return links and timestamps preserved (BR-17); no ledger rows deleted
AST-050	Lost asset keeps last-known condition and location	Boundary	P2	Both	Asset ASSIGNED, condition=GOOD	1. Mark availability=LOST	availabilityState=LOST; conditionGrade retained as last known; currentLocationId/last-known kept; lifecycle stays DEPLOYED (per mapping #16); event logged

3.5 Status-Dimension Filtering & Bulk Operations

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
AST-051	Multi-dimension query answerable (the flagship query)	Functional	P1	Both	Seeded assets across dimensions/locations	1. Filter ownership=LEASED, lifecycle=DEPLOYED, condition=POOR, location=Warehouse-BLR	Returns exactly the leased+deployed+poor+Warehouse-BLR assets; independent AND across all four dimensions; impossible under old single enum, proving the redesign
AST-052	"What can I hand out" fast filter	Functional	P2	Both	Mixed availability assets	1. Filter availability=AVAILABLE	Returns only AVAILABLE assets (backed by partial index); excludes ASSIGNED/RESERVED/IN_TRANSIT/IN_REPAIR/LOST
AST-053	Filtering never crosses tenant boundary	RBAC	P1	API	T1 and T2 both have DEPLOYED leased laptops	1. As T1, run AST-051 filter	Only T1 rows returned; zero T2 rows (BR-03, tenant-first index); count matches T1 seed
AST-054	Empty result for impossible-but-legal combo	Boundary	P3	Both	No asset matches	1. Filter ownership=BYOD, availability=IN_REPAIR, location=Warehouse-DEL	Returns empty set gracefully (no error, no partial merge); pagination shows 0

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
AST-055	Bulk assign to multiple users honors per-asset guardrails	Functional	P2	Web	5 AVAILABLE assets selected; 1 is actually RETIRED	1. Bulk-assign selection	4 assigned (pending acceptance); the RETIRED one rejected individually with reason; partial-success report returned; no all-or-nothing silent drop
AST-056	Bulk condition update appends one ConditionLog per asset	Data-integrity	P2	Both	10 assets selected	1. Bulk set condition=FAIR	10 ConditionLog rows appended (one each); each Asset.conditionGrade cache=FAIR; each version incremented; audit event per asset
AST-057	Bulk operation is transactional per item under contention	Data-integrity	P2	API	Bulk transfer where one item's version is stale	1. Bulk transfer 20 assets, 1 concurrently edited	The stale item returns 409 and is skipped; other 19 succeed; no partial custody corruption; failures itemized in response
AST-058	Bulk retire blocked items reported, not silently skipped	Negative	P2	Web	Bulk retire of 8 assets; 2 still ASSIGNED	1. Bulk retire	6 retired; 2 blocked with "unassign or mark LOST first" (\$1.4); response lists blocked IDs + reason

3.6 RBAC & Mobile-Specific (offline, scan)

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
AST-059	Transfer requires <code>asset:transfer</code> permission	RBAC	P1	Both	User lacks <code>asset:transfer</code>	1. Attempt transfer	Denied (deny-by-default, BR-01); no AssetTransfer row; audit records denied attempt
AST-060	User cannot act on asset outside their data scope	RBAC	P1	Both	User scoped to Site-A; asset at Site-B	1. Attempt assign/return of Site-B asset	Denied by row-level scope filter (BR-02) even though permission string held; asset unchanged
AST-061	Scan-to-assign is a mobile flow; assign/list also work on web (twin)	Functional	P2	Both	Asset AVAILABLE; assignee known	1. Mobile: scan asset → assign to user 2. Web: same assign via detail page	Both surfaces create identical assignment + pending acceptance; scan is the mobile fast-path, web uses detail form; resulting state identical
AST-062	Offline mobile scan queues action, syncs without duplicate	Data-integrity	P2	Mobile	Device offline; asset cached	1. Offline scan + condition update 2. Reconnect 3. Sync	Action queued locally, replayed once on reconnect; server dedupes (idempotency); exactly one ConditionLog appended; version conflict during sync yields 409 handled with reconcile prompt
AST-063	Offline-queued assign conflicts with server-side assign on sync	Negative	P2	Mobile	Offline assign queued; another user assigned same asset online meanwhile	1. Reconnect + sync	Queued assign rejected (409/"already assigned"); user notified; no double custody; local queue cleared for that item

4. License Management & Hardware-Detail — Test Scenarios

4.1 Seat Assignment & the Hard Seat-Limit Block

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
LIC-001	Assign a seat with capacity available	Functional	P1	API	M365 E3, 10 purchased, 3 used, Default Pool	1. POST <code>/licenses/{id}/assignments</code> for a new user	201 Created; <code>seats_used</code> →4, <code>seats_available</code> →6; <code>assigned</code> audit event written; assignee & reconciliation watch notified
LIC-002	Fill the pool exactly to the limit (10th assignment)	Boundary	P1	API	10 purchased, 9 used	1. Assign the 10th distinct user	201 Created; <code>seats_used</code> =10, <code>seats_available</code> =0; no block; low-seat warning state now active
LIC-003	Hard block on the 11th assignment when 10/10 used — transactional block/rollback	Negative	P1	Both	10 purchased, 10 used	1. Attempt to assign an 11th user (web AND mobile)	409 SeatLimitExceeded; guarded <code>UPDATE ... WHERE seats_reserved+1<=seats_allocated</code> affects 0 rows; transaction ROLLBACK; NO assignment row created; <code>seats_used</code> stays 10, <code>seats_reserved</code> unchanged
LIC-004	11th-assignment block — exact UI error modal message	Negative	P1	Both	10/10 used	1. Submit assign dialog on web; 2. repeat on mobile	Modal blocks submit; red banner reads exactly "License Limit Exceeded / Available: 0 / Purchased: 10 / Assigned: 10 / Please purchase additional licenses."; primary CTA becomes "Request additional licenses"; mobile shows the identical block modal
LIC-005	11th-assignment block — dashboard alert card	Functional	P1	Web	10/10 used; dashboard open	1. Trigger a blocked assign; 2. observe Seat-Limit Alerts widget	Live "License Limit Exceeded — M365 E3" alert card appears, fed by <code>assign_blocked</code> , showing count of blocked attempts in last 24h
LIC-006	11th-assignment block — email + in-app notification	Functional	P1	Both	10/10 used; SMTP capture enabled	1. Trigger blocked assign	In-app toast + notification-center entry to acting admin immediately; email to License Admins + license owner, subject "Seat limit reached — Microsoft 365 E3 (0 of 10 available)" with Buy/Request-more deep link
LIC-007	11th-assignment block — immutable <code>assign_blocked</code> audit event	Security	P1	API	10/10 used	1. Trigger blocked assign; 2. read <code>/licenses/{id}/history</code> ; 3. attempt to UPDATE/DELETE the event	<code>LicenseAuditEvent{event_type:"assign_blocked", reason:"License Limit Exceeded", subject_ref:"user:<id>", after:{purchased:10,used:10,available:0}}</code> present; event is append-only — update/delete rejected
LIC-008	All five block reactions fire together (aggregate E2E)	E2E	P1	Both	10/10 used	1. From UI attempt the 11th assign	Single blocked attempt produces ALL five: (a) transactional block/rollback, (b) UI error modal with exact message, (c) dashboard alert card, (d) email + in-app notification, (e) immutable <code>assign_blocked</code> audit — atomically and consistently
LIC-009	409 block response body contract	Negative	P1	API	10/10 used	1. POST assignment	Body matches contract: <code>error:"SeatLimitExceeded"</code> , <code>message</code> string, <code>details:{purchased:10,assigned:10,available:0,licenseId},remediation:{action:"request_additional",url}</code>
LIC-010	Race condition — two admins grab the last seat simultaneously	Concurrency	P1	API	10 purchased, 9 used (1 free seat)	1. Fire two concurrent POST assignments (distinct users) at the same pool	Exactly one succeeds (201), the other gets 409 SeatLimitExceeded; final <code>seats_used</code> =10; no over-commit; one <code>assigned</code> + one <code>assign_blocked</code> event; SERIALIZABLE row lock serializes writers
LIC-011	Race condition — no seat leak / no double-count under load	Concurrency	P1	API	10 purchased, 5 used	1. Fire 20 concurrent distinct-user assignments	Exactly 5 succeed, 15 blocked; <code>seats_used</code> =10, <code>seats_reserved</code> =10; roll-up <code>SUM(seat_pool.seats_reserved)</code> equals assignment count; invariant <code>0 ≤ seats_reserved ≤ seats_purchased</code> holds
LIC-012	Low-seat (≤10%) warning badge	Boundary	P2	Web	100 purchased, 90 used (10% remaining)	1. Open license row and assign dialog	Amber warning badge "Only 10 seats remaining." shown on row and dialog; assignment still permitted
LIC-013	Low-seat warning boundary crossing at 9%	Boundary	P2	Web	100 purchased, 91 used	1. View row	Amber "Only 9 seats remaining" persists (≤10% threshold inclusive); no block until 0 available
LIC-014	Low-seat digest email to License Admin	Functional	P3	API	≤10% seats remaining	1. Cross the low-seat threshold	In-app alert immediate; email coalesced into digest per notifications matrix
LIC-015	Duplicate active assignment for same principal blocked (V4)	Negative	P1	API	User U already active on license; seat free	1. POST assignment for U again	409/DuplicateAssignmentError; ROLLBACK; unique index <code>(license_id,user_id) WHERE status='active'</code> prevents double-consume; <code>seats_used</code> unchanged
LIC-016	Reassign a principal after revoke succeeds	Functional	P2	API	U revoked earlier; seat free	1. POST assignment for U	201 Created; prior revoked row does not block; new active row bound to fresh <code>reservation_token</code>

4.2 Bulk, Transfer, Revoke, Reclaim

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
LIC-017	Bulk assign partial-success (assign up to limit, rest blocked)	Boundary	P1	API	10 purchased, 3 used (7 free); CSV of 10 principals	1. POST <code>/assignments:bulk</code> with 10 rows	Report <code>{assigned:7, blocked:3}</code> ; first 7 reserved, last 3 each carry per-row reason "License Limit Exceeded"; <code>seats_used</code> = 10; single <code>bulk_assigned</code> audit with child manifest
LIC-018	Bulk assign all-or-nothing rollback mode	Negative	P2	API	10 purchased, 5 used; bulk of 8; mode=atomic	1. POST bulk with atomic flag	Insufficient capacity → entire batch ROLLBACK; 0 assigned; <code>seats_used</code> = 5 unchanged; report explains all-or-nothing rejection
LIC-019	Bulk assign with a duplicate row in payload	Negative	P2	API	Capacity available; CSV contains same user twice	1. POST bulk	First occurrence assigned; duplicate row blocked with V4 reason; no double seat consumed
LIC-020	Bulk remove releases seats via guarded decrements	Functional	P2	API	10 used	1. POST <code>/assignments:bulkRemove</code> for 4 assignments	4 seats released; <code>seats_used</code> -6; guarded <code>seats_reserved = seats_reserved - 1 WHERE seats_reserved > 0</code> ; one <code>bulk_removed</code> audit with child manifest
LIC-021	Revoke a single seat releases capacity	Functional	P1	Both	10/10 used	1. DELETE <code>/assignments/{aid}</code> ; 2. re-attempt an assign	Assignment status → revoked, <code>revoked_at</code> set, <code>seats_used</code> -9; the previously-blocked new assign now succeeds
LIC-022	user → user transfer is net-zero and atomic	Functional	P1	API	Source user active; license unit=user	1. POST <code>/transfer</code> source → target	Source revoked (release), target created (re-reserve) in one transaction; net seat delta 0; <code>transferred</code> status recorded; <code>transferred</code> audit event
LIC-023	device → device transfer with key re-binding	Functional	P2	API	Key-bound (Windows) license; source device active	1. POST device → device transfer	Old key deactivated, new activated respecting <code>max_activations</code> ; net-zero seats; atomic; <code>activations_used</code> consistent
LIC-024	Transfer target reservation fails — source restored, no seat leak	Negative	P1	API	Target invalid / would exceed a different pool	1. POST transfer expecting target failure	Whole transaction ROLLBACK; source assignment remains active; no seat released-then-lost; seat counters unchanged
LIC-025	Cross-unit user → device transfer blocked when family disallows (V3)	Negative	P2	API	License <code>unit_of_assignment=user</code> only	1. POST user → device transfer	Blocked with validation V3 (principal type mismatch); no state change
LIC-026	Idle-seat reclamation queue population	Functional	P2	API	User seat active, <code>last_usage_at</code> > 60 days; device seat offline > 60 days	1. Run unused-detection job; 2. GET <code>/reports/unused</code>	Both flagged as reclaimable with recovered-cost estimate; appear in reclamation queue
LIC-027	Bulk reclaim from queue releases seats	Functional	P2	Both	Reclamation queue has 5 idle seats	1. Select all; 2. one-click bulk reclaim	5 seats released (guarded decrement); <code>reclaimed</code> audit events; unused-spend-recovery report updates
LIC-028	Reclaim does not touch recently-active seats	Negative	P2	API	Mixed seats, some active within window	1. Run detection	Only seats past staleness threshold flagged; active-within-window seats excluded from queue

4.3 Renewal, Expiry, Duplicate/Over-Purchase, Keys, Validation

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
LIC-029	Renewal reminder at 90-day threshold	Functional	P2	API	Subscription license, renewal_date exactly 90d out	1. Run daily scheduler	Reminder fires; manual license → action-required task to Procurement; in-app + email per matrix
LIC-030	Renewal reminders at 60/30/7-day thresholds	Boundary	P2	Both	renewal_date at 60, 30, 7 days out (three fixtures)	1. Run scheduler on each date	A reminder fires at each threshold exactly once; no duplicates on non-threshold days
LIC-031	Day-0 (on-expiry) reminder + auto-renewal heads-up	Functional	P2	API	One auto-renewal, one manual, both at day-0	1. Run scheduler	Manual → action-required; auto → "will auto-renew on <date> at ~<cost>" heads-up to Admin/Finance
LIC-032	Expiry suspends dependent assignments	Functional	P1	API	Subscription license past <code>expiry_date</code> , seats active	1. Run daily expiry job	Status → Expired; dependent assignments suspended (per grace config); <code>expired</code> audit; alerts raised; suspended seats no longer count as active-consuming
LIC-033	Cannot assign against Expired/Retired license (V8)	Negative	P1	API	License status=Expired	1. POST assignment	Rejected by V8; no reservation; clear error
LIC-034	Renewal completion rolls expiry + seats forward	Functional	P2	API	Planned renewal with <code>seats_after</code> > <code>seats_before</code>	1. POST <code>/renewals/{rid}:complete</code>	New <code>expiry_date</code> applied; seats trued-up; status → Active; <code>renewed</code> audit with seat delta
LIC-035	Duplicate license-key fingerprint detection (warn, not block)	Functional	P2	API	Two licenses share identical normalized key	1. Add key to 2nd license; 2. GET <code>/reports/duplicates</code>	V10 raises warning (not block); pair surfaces in duplicate queue by <code>key_fingerprint</code> ; key never exposed

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
LIC-036	Over-purchase detection (same product+edition, idle seats)	Functional	P3	API	Two M365 E3 licenses, one with many idle seats	1. GET <code>/reports/duplicates</code>	Over-purchase flagged with idle-seat count; consolidation report suggests reduction
LIC-037	Same principal holding overlapping entitlements flagged	Functional	P3	API	User active on two overlapping licenses	1. Run duplicate detection	Overlap surfaced as warning; no auto-revoke
LIC-038	Per-family unit-of-assignment validation user vs device (V3)	Negative	P1	API	Windows license unit=device	1. POST assignment with <code>assignee_type=user</code>	Rejected by V3; error names required unit=device; no seat consumed
LIC-039	Key-bound family requires key with remaining activations (V5)	Negative	P2	API	Windows license, no key or exhausted key	1. POST assignment without valid <code>license_key_id</code>	Rejected by V5; blocked until a key with remaining activations supplied
LIC-040	License-key reveal is permissioned + audited	Security	P1	API	Actor lacks <code>license.key.reveal</code> ; then Admin with permission	1. POST <code>/keys/{keyId}:reveal</code> as unauthorized; 2. retry as Admin	Unauthorized → 403, no plaintext; authorized → key returned once and mandatory audit event written; list APIs never return plaintext
LIC-041	Tenant isolation on assignment	Security	P1	API	Two tenants each own a license	1. Tenant A actor POSTs assignment to Tenant B's license id	403/404 (row-level security); no cross-tenant seat mutation; audit records the denied attempt
LIC-042	Seats-purchased cannot go below seats-used on adjustment	Boundary	P2	API	10 purchased, 8 used	1. PATCH <code>seats_purchased</code> = 6	Rejected — would violate invariant <code>seats_used ≤ seats_purchased</code> (V1); counters unchanged
LIC-043	Perpetual vs subscription expiry validation (V6/V7)	Negative	P3	API	Attempt perpetual with expiry set; subscription with null expiry	1. POST each	Perpetual requires null <code>expiry_date</code> ; subscription requires non-null; both invalid inputs rejected; <code>expiry_date ≥ purchase_date</code> enforced
LIC-044	Mobile assign/revoke enforces identical server guard	E2E	P1	Mobile	10/10 used; native app online	1. Attempt assign on mobile; 2. revoke one; 3. assign again	11th blocked with same modal + 409; after revoke, assign succeeds; offline mutation refused (reservation must be transactional server-side)

4.4 Hardware / Asset-Detail 7-Tab Page

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
HW-001	System Information tab renders with source badges	Functional	P2	Web	Asset with Agent + Manual + Intune data	1. Open asset; 2. select System tab	All fields render; each carries a source badge (Manual/Agent/Intune/Discovery/Entra) and last-synced timestamp
HW-002	OS tab renders + BitLocker key never shown inline	Security	P1	Both	Asset with OS + SecurityPosture data	1. Open OS tab	OS fields render with badges; BitLocker Recovery Key shows escrow-present yes/no only — raw key never displayed (permissioned reveal elsewhere); Windows key masked
HW-003	Hardware tab renders full inventory with freshness	Functional	P2	Web	Agent-collected hardware data	1. Open Hardware tab	CPU/RAM/disk/battery/network render; SMART, battery health %, cycle count shown with per-field <code>collected_at</code>
HW-004	Installed Software tab reconciliation states	Functional	P2	Web	Apps with mixed entitlement states	1. Open Installed Software tab	Each entry reconciled: licensed / unlicensed / unmanaged; License Link FK resolves to <code>SoftwareLicense</code>
HW-005	Microsoft 365 tab renders via Graph with badges	Functional	P2	Both	M365Profile populated from Graph	1. Open M365 tab	Assigned license, Exchange/OneDrive/Teams, MFA, Last Sign-In shown with Entra/Graph badges; license links to Part A
HW-006	Warranty tab renders + remaining-days derived	Functional	P2	Web	WarrantyContract with start/end dates	1. Open Warranty tab	Vendor, dates, derived Remaining Days shown; attachments/invoice links present; source badges Manual/Vendor API
HW-007	Stale-data indicator when field past staleness window	Boundary	P2	Web	A field's <code>collected_at</code> older than its staleness window	1. Open the relevant tab	Field flagged "stale"; badge shows age; fresher-tier value preferred where available per merge rules
HW-008	Source-precedence: security field prefers Intune/Agent over Manual	Functional	P3	API	BitLocker value differs between Manual and Intune	1. Ingest both; 2. read field	Authoritative-source-wins: Intune/Agent value surfaces; Manual overridden; badge reflects winning source + confidence
HW-009	Manual-override lock on procurement fields	Functional	P3	API	Asset tag set Manual; discovery reports different	1. Ingest discovery	Manual value retained (override lock); discovery only fills when Manual empty
HW-010	Asset Health score = weighted average of sub-scores	Functional	P1	Both	Sub-scores Battery/Storage/Memory/Warranty/Security/Updates all ≥60	1. Open Asset Health tab	Overall = round($\sum \text{subScore} \times \text{weight}$) with weights 15/20/10/15/25/15; grade band matches computed value
HW-011	Capping rule — Security < 40 caps overall at "Poor"	Boundary	P1	Both	High sub-scores but Security=35 (unencrypted disk)	1. Compute health	Weighted avg may be ≥75 but overall capped at min(overall,59)=Poor; grade shows "Poor" not "Good/Excellent"

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
HW-012	Capping rule — Storage < 40 caps overall at "Poor"	Boundary	P1	Both	Failing SMART → Storage=0, other sub-scores high	1. Compute health	Overall capped at 59 "Poor" regardless of weighted average; safety-critical dimension dominates
HW-013	No cap when both Security and Storage ≥ 40	Boundary	P2	API	Security=45, Storage=42, others high	1. Compute health	No cap applied; overall = true weighted average; grade can be Good/Excellent
HW-014	Grade band boundaries	Boundary	P3	API	Fixtures at 89, 90, 74, 75, 59, 60, 40, 39	1. Compute each	Bands resolve exactly: 90–100 Excellent, 75–89 Good, 60–74 Fair, 40–59 Poor, <40 Critical
HW-015	Recommendations engine output from sub-scores	Functional	P2	Both	Storage<50 & free<10%; Battery<50; BitLocker off; Warranty<30d; unsupported OS	1. Open Asset Health tab	Prioritized items generated, e.g. disk replacement, battery replacement, enable BitLocker, renew AMC, schedule OS upgrade — each tied to its failing sub-score
HW-016	Health recomputes on fresh ingestion	E2E	P3	API	Health=Poor due to BitLocker off	1. Ingest Agent update showing BitLocker on; 2. reopen Asset Health	Security sub-score recovers; cap lifts; overall grade recomputes upward; badge shows new source/collected_at
HW-017	Cross-module link: Installed Software → License seat usage	E2E	P2	Web	Installed M365 reconciled to a Part A license at 10/10	1. From Installed Software click License Link	Navigates to license showing seats 10/10; consistent with LIC-003 block state

5. Procurement & Inventory — Test Scenarios

5.1 Purchase Requests (PR)

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
PRC-001	Create PR from approved asset request (auto-populated)	Functional	P1	Web	Approved asset request exists; active cost-center	1. Open approved asset request 2. Create PR from it 3. Save	PR created with <code>source_type=asset_request</code> , <code>asset_request_id</code> linked, lines/qty/UoM auto-filled from request; <code>pr_number</code> matches <code>PR-{YYYY}-{seq}</code>
PRC-002	Create standalone PR	Functional	P2	Web	User has create-PR right; active cost-center	1. New PR, <code>source_type=standalone</code> 2. Add lines, <code>needed_by</code> ≥ today 3. Submit	PR persisted with status transitioning draft→submitted; no <code>asset_request_id</code> required
PRC-003	PR asset-request source without <code>asset_request_id</code>	Negative	P2	API	—	1. POST PR with <code>source_type=asset_request</code> and null <code>asset_request_id</code>	422 RFC-7807; validation error " <code>asset_request_id</code> required when <code>source_type=asset_request</code> "
PRC-004	PR <code>needed_by</code> in the past	Negative	P2	API	—	1. POST PR with <code>needed_by</code> = yesterday	422; <code>needed_by</code> must be ≥ today; PR not created
PRC-005	PR with inactive cost-center	Negative	P2	API	Cost-center <code>status=inactive</code>	1. POST PR referencing inactive <code>cost_center_id</code>	422; <code>cost_center</code> must be active; blocked
PRC-006	PR line quantity = 0	Boundary	P2	API	—	1. POST PR line with <code>quantity=0</code>	422; quantity must be > 0
PRC-007	PR over policy threshold without justification	Negative	P2	Web	Tenant policy sets justification threshold	1. Create PR with <code>estimated_total</code> above threshold, blank justification 2. Submit	Submit blocked; justification required
PRC-008	Attachment exceeds 25MB / disallowed MIME	Negative	P3	Web	—	1. Attach 30MB .exe to PR	Upload rejected; size and MIME allow-list enforced

5.2 RFQ, Quotation & Award

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
PRC-010	Issue RFQ to N active vendors	Functional	P1	Web	Approved PR; ≥ 2 active vendors	1. Create RFQ from PR 2. Add 3 vendors 3. Send	RFQ status draft→sent; <code>rfq_number</code> generated; invited vendors notified
PRC-011	RFQ with zero vendors	Negative	P2	API	Approved PR	1. POST RFQ with empty <code>vendor_ids</code>	422; ≥ 1 active vendor required
PRC-012	RFQ includes blacklisted/inactive vendor	Negative	P1	API	One vendor blacklisted	1. POST RFQ with blacklisted <code>vendor_id</code>	422; blacklisted/inactive vendor cannot receive RFQ
PRC-013	RFQ <code>due_date</code> not in future	Boundary	P2	API	—	1. POST RFQ with <code>due_date</code> = now	422; <code>due_date</code> must be > now
PRC-014	Vendor submits quote via portal	Functional	P2	Web	RFQ sent; vendor invited	1. Vendor logs into portal 2. Enters per-line price, lead time, warranty, <code>valid_until</code> 3. Submit	Quote captured; one quote per vendor per RFQ; visible to procurement officer
PRC-015	Weighted scoring computed correctly	Functional	P1	API	3 submitted quotes with varied <code>price/lead/rating/warranty</code>	1. GET comparison matrix	Each quote scored 0–100 using default weights (0.40 price + 0.20 lead + 0.20 rating + 0.10 warranty + 0.10 compliance); lowest price/shortest lead score highest; auto-recommendation = top score
PRC-016	Expired quote excluded from scoring	Negative	P2	API	Quote <code>valid_until</code> < today	1. GET comparison	Expired quote excluded from eligibility/scoring; not recommended
PRC-017	Award top-scored quote	Functional	P1	Web	Scored comparison exists	1. Award recommended quote	Quote <code>is_selected=true</code> ; RFQ closed; no reason required
PRC-018	Award non-top-scored quote without reason	Negative	P1	API	Comparison with clear top score	1. POST award of lower-scored quote, no reason	422; <code>selection_reason</code> required for override
PRC-019	Award non-top-scored quote with override reason	Functional	P2	Web	Comparison exists	1. Award lower quote with reason text	Award accepted; <code>selection_reason</code> persisted; audit-logged

5.3 Purchase Orders — Budget, Threshold, Vendor Guards

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
PRC-020	PO from winning quote inherits traceability	Functional	P1	Web	Awarded quotation	1. Create PO from award	PO carries <code>purchase_request_id</code> + <code>quotation_id</code> ; lines/prices inherited
PRC-021	Standalone PO without elevated permission	Negative	P2	API	User lacks elevated right	1. POST PO with no PR/quote	403/422; standalone PO requires elevated permission + justification
PRC-022	Budget HARD gate blocks insufficient budget	Negative	P1	Both	Budget <code>amount_available</code> < <code>PO.total</code>	1. Submit PO for approval 2. Budget check runs	PO blocked (PO_Rejected/held); overspend reason returned; routed to budget-owner escalation; no hard commitment created

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
PRC-023	Budget check passes at exact availability	Boundary	P1	API	amount_available == PO.total	1. Submit PO 2. Budget check	Passes (available ≥ total); PO proceeds to approval
PRC-024	Budget available one cent short	Boundary	P1	API	available = total - 0.01	1. Submit PO	Blocked; insufficient budget
PRC-025	Approval auto-band (< 1,000)	Functional	P1	API	Threshold matrix default	1. Submit PO total 900	approval_state=not_required/auto-approved; no approver routing
PRC-026	Approval manager band (1,000–10,000)	Functional	P1	Both	Manager approver configured	1. Submit PO total 5,000	Routed to Procurement Manager; approval_state=pending
PRC-027	Approval dept-head band (10,000–50,000)	Functional	P1	Both	Dept head configured	1. Submit PO total 30,000	Routed to department head
PRC-028	Approval CFO band (> 50,000, parallel)	Functional	P1	Both	CFO + procurement lead configured	1. Submit PO total 75,000	Parallel approval to CFO + procurement lead; both required
PRC-029	Band boundary at exactly 1,000 / 10,000 / 50,000	Boundary	P1	API	Matrix default bands	1. Submit POs of 1000, 10000, 50000	Each routes to correct band edge per tenant band definition; deterministic and documented
PRC-030	Revision increasing total re-triggers approval	Functional	P1	Both	Approved PO total 900 (auto)	1. Revise PO up to 6,000 2. Save	revision_no++; approval recalculated to manager band; re-enters pending approval
PRC-031	Revision not increasing total keeps approval	Boundary	P2	API	Approved PO	1. Revise PO with lower/equal total	No re-approval required; version history recorded
PRC-032	PO to blacklisted vendor blocked	Negative	P1	API	Vendor status=blacklisted	1. POST PO with blacklisted vendor_id	422; blacklisted vendor cannot receive PO
PRC-033	Contract-required category without active contract	Negative	P1	Both	Category flagged contract-required; vendor has no active contract	1. Issue PO for that category	Blocked; "active contract required for category"
PRC-034	Contract-required category with valid contract	Functional	P2	Web	Active vendor contract covering category	1. Issue PO	PO issued successfully
PRC-035	Mobile one-tap PO approve with reason	Functional	P1	Mobile	Pending PO push notification	1. Open push 2. View budget impact + attachments 3. Approve with note	PO approved; approver + reason captured; soft→hard commitment converted
PRC-036	Mobile PO reject	Functional	P2	Mobile	Pending PO	1. Reject with reason	approval_state=rejected; requester notified; commitment released
PRC-037	Approving PO converts soft→hard commitment	Data-integrity	P1	API	PR soft commitment exists	1. Approve PO	budget.amount_committed reflects hard commitment; PR soft commit superseded; amount_available recomputed

5.4 Goods Receipt (GRN) & QC

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
PRC-040	Full receipt all lines accepted	Functional	P1	Both	PO acknowledged/shipped	1. Create GRN, receive full qty 2. QC pass 3. Post	GRN posted; PO line quantity_received = ordered; emits inventory.stock.received
PRC-041	Partial receipt leaves PO line open	Functional	P1	Both	PO with qty 100	1. Receive 60, QC pass, post	receipt_type=partial; quantity_received=60; PO line remains open for remaining 40
PRC-042	Over-receipt beyond tolerance blocked	Negative	P1	Both	over_receipt_tolerance_pct=5%; ordered 100	1. Attempt receive 110	Blocked; exceeds remaining × (1+tol); GRN not posted
PRC-043	Over-receipt within tolerance accepted	Boundary	P2	API	tol=5%; ordered 100	1. Receive 105	Accepted (= remaining × 1.05 cap)
PRC-044	Rejected qty routed to quarantine	Data-integrity	P1	Both	PO shipped	1. Receive 100, accept 90, reject 10, QC partial 2. Post	90 → receipt (Available), 10 → quarantine movement; rejected qty NEVER enters available
PRC-045	quantity_accepted > quantity_received	Negative	P2	API	—	1. Post GRN line accept 50 of received 40	422; accepted ≤ received
PRC-046	Serial count mismatch for serialized item	Negative	P1	Both	Serialized stock item; accept 5	1. Post GRN with 3 serials	422; serial count must equal accepted qty
PRC-047	GRN warehouse ≠ PO ship_to	Negative	P2	API	—	1. Post GRN with different warehouse_id	422; warehouse must equal PO ship_to
PRC-048	Mobile receive with barcode/QR scan	Functional	P1	Mobile	PO shipped; mobile receiver	1. Scan packing slip barcode 2. Scan serials 3. Photo damage for rejects 4. Post	Lines matched by scan; serials captured; damage photo attached to quarantine; GRN posted
PRC-049	GRN post increments budget actual	Data-integrity	P1	API	Posted GRN	1. Post GRN	budget.amount_actual incremented; valuation recomputed; movement reference_id = GRN id

5.5 3-Way Match, Payment & Closure

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
PRC-050	3-way match pass within tolerance	Functional	P1	Web	PO + GRN accepted + invoice aligned	1. POST invoice match	match_status=matched; qty_variance & price_variance within tolerance for every line; payment allowed

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
PRC-051	Match exception on qty variance	Negative	P1	Both	Invoice qty > GRN accepted qty beyond tolerance	1. Run match	match_status=exception; payment blocked; procurement + finance notified
PRC-052	Match exception on price variance	Negative	P1	Both	Invoice unit price > PO price beyond tolerance	1. Run match	match_status=exception; payment blocked pending resolve/override
PRC-053	Variance exactly at tolerance edge	Boundary	P2	API	price_variance == tolerance_pct	1. Run match	Passes (
PRC-054	Payment guard blocks pay while unmatched	Negative	P1	Both	Invoice match_status=unmatched	1. Attempt approve payment	Blocked; no payment approved while status €
PRC-055	Payment blocked while exception	Negative	P1	Both	match_status=exception	1. Attempt payment	Blocked until resolved or overridden
PRC-056	Finance-only override of match exception	Functional	P1	Both	Exception; user has finance role	1. POST match/override with reason	match_status=overridden; reason + finance actor logged; payment unblocked
PRC-057	Non-finance role cannot override	Negative	P1	API	Exception; procurement officer (no finance role)	1. POST match/override	403; override requires finance role
PRC-058	2-way match fallback for service line	Functional	P2	API	Service PO, no GRN	1. Run match	match.type=two_way (PO→Invoice only); passes without GRN
PRC-059	Sum of partial payments cannot exceed invoice total	Boundary	P1	API	Invoice total 1,000; 800 already paid	1. POST payment 300	Blocked; Σ payments $\leq$ invoice.total; max 200 accepted
PRC-060	Partial payment within total	Functional	P2	Web	Matched invoice	1. Pay 400 of 1,000	payment.status=partial; paid_amount=400
PRC-061	PO auto-closes when fully received/invoiced/paid	E2E	P1	API	All lines received, invoiced, reconciled	1. Complete final payment reconciliation	PO status auto → closed; residual commitment zero
PRC-062	Manual short-close with reason	Functional	P2	Web	PO partially received	1. Short-close PO with reason	PO closed; reason logged; residual commitment reversed/released
PRC-063	FX variance beyond tolerance is a match exception	Negative	P1	API	PO currency EUR, invoice USD; FX drift beyond tolerance	1. Run match at captured fx_rate	Cross-currency compared at fx_rate; FX variance out of tolerance → exception; payment blocked
PRC-064	Cross-currency match within FX tolerance	Functional	P2	API	PO/invoice different currency, minor drift	1. Run match	Converted at captured fx_rate; within tolerance → matched
PRC-065	End-to-end source-to-pay happy path	E2E	P1	Both	Approved asset request	1. PR → RFQ → award → PO(budget+approval) → ship → GRN → match → pay → close	Every stage transitions per lifecycle; stock incremented; asset chain traceable; PO_Closed reached

5.6 Vendor Management

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
PRC-070	Create vendor with duplicate legal_name	Negative	P2	API	Vendor legal_name exists in tenant	1. POST vendor same legal_name	409/422; unique legal_name per tenant enforced
PRC-071	Blacklist vendor blocks new POs	Functional	P1	Web	Active vendor with open POs	1. Blacklist vendor	status=blacklisted; new RFQ/PO issue blocked; existing POs unaffected
PRC-072	Contract expiry renewal alert	Functional	P3	Web	Contract end_date within renewal_alert_days	1. Run alert job	Vendor manager notified of expiring contract
PRC-073	Vendor bank details encrypted at rest	Data-integrity	P2	API	Vendor with bank_details	1. Store bank details 2. Inspect storage	bank_details stored encrypted; not returned in plaintext to unauthorized roles
PRC-074	Vendor portal user restricted to own records	Negative	P1	Web	External vendor identity	1. Attempt to view another vendor's RFQ	403; portal user scoped to own vendor + invited RFQs only

5.7 Inventory — Locations & Invariants

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
INV-001	Non-negative invariant on issue	Data-integrity	P1	API	on_hand=10	1. Issue 15	Rejected atomically; on_hand cannot go < 0; no movement written
INV-002	available = on_hand – reserved always holds	Data-integrity	P1	API	on_hand=100, reserved=30	1. GET stock level	qty_available=70 exactly; computed invariant
INV-003	reserved cannot exceed on_hand	Boundary	P1	API	on_hand=50, reserved=50	1. Reserve 1 more	Rejected; $0 \leq$ reserved $\leq$ on_hand
INV-004	Location hierarchy cycle rejected	Negative	P2	API	Location A parent of B	1. Set A.parent_id=B	422; parent chain must be acyclic

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
INV-005	Delete location holding stock	Negative	P2	API	Location with on_hand>0	1. DELETE location	Blocked; soft-delete only; location with stock cannot be removed
INV-006	Put-away to non-receivable bin	Negative	P2	API	Bin is_receivable=false	1. Put-away stock there	Blocked; stock only rests in receivable bins
INV-007	Issue from non-pickable bin	Negative	P2	API	Bin is_pickable=false	1. Issue from bin	Blocked; not pickable

5.8 Inventory — Reservation, Issue, FEFO/FIFO

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
INV-010	Reserve decreases available not on_hand	Functional	P1	API	on_hand=100	1. Reserve 20	reserved=20; available=80; on_hand unchanged=100; reservation movement logged
INV-011	Release reservation restores available	Functional	P1	API	reserved=20	1. Release reservation	reserved=0; available=100; release movement logged
INV-012	Issue converts reserved→out, reduces on_hand	Data-integrity	P1	API	reserved=20, on_hand=100	1. Issue reserved 20	on_hand=80; reserved=0; issue movement; ledger balanced
INV-013	Fractional issue allowed for consumable	Functional	P1	Both	item_type=consumable, on_hand=5.0 L	1. Issue 1.5 L to cost-center	Accepted; on_hand=3.5; consumption tagged to cost-center
INV-014	Fractional issue blocked for durable/serialized	Negative	P1	API	item_type=durable serialized	1. Issue 0.5 unit	422; durable/serialized require whole-unit movement
INV-015	FEFO picking suggests earliest expiry	Functional	P2	Both	Two batches, differing expiry	1. Request pick	Earliest-expiry batch suggested first
INV-016	Expired batch blocks issue	Negative	P1	Both	Batch expiry < today	1. Attempt issue from expired batch	Blocked; routed to disposition; cannot issue
INV-017	FIFO picking by cost policy	Functional	P3	API	FIFO policy, non-batch item	1. Request pick	Oldest-cost layer suggested first

5.9 Inventory — Quarantine, Reorder, Transfers

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
INV-020	Quarantine stock cannot be reserved/issued/deployed	Data-integrity	P1	API	qty_quarantine=10, available=0	1. Attempt reserve/issue/deploy from quarantine	All blocked; quarantine excluded from available until disposition
INV-021	Quarantine disposition return-to-stock	Functional	P2	Web	Quarantined qty, repaired	1. POST disposition action=return	Qty moves quarantine→available; movement logged
INV-022	Quarantine disposition scrap	Functional	P2	Web	Quarantined qty	1. Disposition action=scrap	disposal movement; qty removed; not available
INV-023	Reorder trigger creates draft PR	Functional	P1	API	available drops to ≤ reorder_point	1. Issue stock crossing reorder point	Low-stock alert raised; draft Purchase Request suggested for reorder_quantity; feeds Procurement
INV-024	Reorder at exact reorder_point boundary	Boundary	P2	API	available == reorder_point	1. GET reorder suggestions	Triggered (available ≤ reorder_point); suggestion present
INV-025	Create PR from reorder suggestion	Functional	P2	Web	Reorder suggestion exists	1. POST create-pr from suggestion	Draft PR created with reorder_quantity; linked to suggestion
INV-026	Transfer atomicity: dispatch reserves & moves to in-transit	Functional	P1	Both	available=100 at source	1. Create transfer qty 40 2. Dispatch	Source reserved then moved to in_transit; transfer_out movement; source available=60
INV-027	Transfer receive moves in-transit→destination	Functional	P1	Both	Dispatched transfer in_transit	1. Receive full 40	Destination on_hand +40; in_transit cleared; transfer_in movement; status=received
INV-028	Transfer short-receipt creates variance	Negative	P1	Both	Dispatched 40	1. Receive 35	5-unit shortage variance flagged; reason + adjustment required; not silently lost
INV-029	Transfer qty exceeds available at source	Negative	P2	API	available=30	1. Dispatch transfer of 40	Blocked; qty ≤ available at source
INV-030	Transfer same source and destination	Negative	P3	API	—	1. Create transfer from=to	422; from and to must differ

5.10 Inventory — Cycle Count & Offline Mobile

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
INV-040	Cycle count posting writes variance movements	Data-integrity	P1	Both	Count with variance; within tolerance	1. Count, review, post	count_adjust movements written per variance; levels updated in one transaction
INV-041	Variance beyond tolerance requires approval	Negative	P1	Both	Variance > tenant tolerance	1. Attempt post	Post blocked pending supervisor approval; not posted until approved
INV-042	Variance reason required on post	Negative	P2	API	Count line variance ≠ 0	1. Post without variance_reason	422; variance_reason required when variance ≠ 0
INV-043	Blind count hides expected qty	Functional	P2	Mobile	Audit is_blind=true	1. Open count sheet	expected_qty hidden from counter; only counted_qty entry shown
INV-044	Offline count sheet download and count	Functional	P1	Mobile	Planned audit; device offline	1. Download sync-package 2. Disconnect 3. Scan + enter quantities	Entries queued locally with scanned_at; no server call required while offline
INV-045	Offline sync on reconnect	Functional	P1	Mobile	Offline entries queued	1. Reconnect 2. Sync count-session	Entries uploaded; server reconciles snapshot version; synced_at set

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
INV-046	Offline sync conflict detected (no last-write-wins)	Data-integrity	P1	Mobile	Concurrent server change since snapshot	1. Sync stale offline entries	conflict_status=detected; last-write-wins disallowed; blocked from posting until reviewer resolves
INV-047	Reviewer resolves conflict then posts	Functional	P2	Both	conflict_status=detected	1. Reviewer resolves 2. Post	conflict_status=resolved; count posts with reconciled values
INV-048	Ledger immutability — no edit/delete	Data-integrity	P1	API	Existing stock_movement	1. Attempt PATCH/DELETE movement	Rejected; append-only; corrections must be new reversing movements

5.11 Inventory — GRN→Stock & Stock→Asset Deployment

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
INV-050	GRN post creates receipt + quarantine movements	E2E	P1	API	GRN posted with accepts + rejects	1. Post GRN (emits <code>inventory.stock.received</code>)	Accepted qty → <code>receipt</code> (Available) with batch/serial; rejected → <code>quarantine</code> ; GRN id = reference_id on each movement
INV-051	Deploy serialized durable unit creates asset + QR	E2E	P1	Both	1 available serialized unit	1. POST deploy	Exactly one <code>deployment</code> movement (on_hand-1); Asset created inheriting SKU/category/unit_cost/serial/PO-GRN provenance; globally-unique QR generated; emits <code>asset.created.from_inventory</code>
INV-052	Non-serialized consumable cannot convert to asset	Negative	P1	Both	Consumable non-serialized item	1. Attempt deploy-to-asset	Blocked; non-serialized consumables cannot become assets
INV-053	Deploy consumes exactly one unit (no over-consume)	Boundary	P1	API	available=1 serialized	1. Deploy unit	on_hand decremented by exactly 1; second deploy of same serial fails (no unit left)
INV-054	Fractional/serial-less deploy rejected	Negative	P2	API	Serialized item	1. Deploy qty 0.5 or missing serial	422; validate one available/reserved unit for that serial; fractional deploy rejected
INV-055	Mobile deploy-to-Asset with scan + QR print	Functional	P1	Mobile	Serialized unit in stock; mobile asset manager	1. Scan stock unit 2. Capture assignee/location 3. Confirm deploy	Asset created; QR label generated and printable/displayable on device; one unit consumed
INV-056	QR uniqueness at asset conversion	Data-integrity	P2	API	Two deployments	1. Deploy two units	Each generates a distinct globally-unique QR; no collision; item/bin/asset barcodes unique per tenant
INV-057	Deploy reserved unit converts reserved→deployed	Data-integrity	P2	API	Unit reserved for deployment	1. Deploy reserved serial	Reserved→Deployed; on_hand-1; reserved decremented; ledger balanced

6. Maintenance, Approvals, Notifications & Audit — Test Scenarios

6.1 Work-Order State Machine (MNT)

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
MNT-001	Reported → Triage transition	Functional	P1	Both	WO created via portal, status=Reported	1. Open WO 2. Submit triage with priority/severity	Status=Triage; triaged_by set; response SLA clock running
MNT-002	Triage → Diagnosis transition	Functional	P1	Web	WO in Triage, fault verified	1. Complete inspection 2. Begin diagnosis	Status=Diagnosis; SLA still running
MNT-003	Diagnosis → Assigned auto-approve (cost ≤ tier-0)	Functional	P1	API	Diagnosis complete, estimated_cost below tier-0 threshold	1. POST diagnosis with estimate 2. Trigger transition	PendingApproval skipped; status=Assigned; approved_cost auto-set
MNT-004	Diagnosis → PendingApproval (cost > threshold)	Functional	P1	Both	estimated_cost above tier-0	1. Submit diagnosis estimate 2. Transition	Status=PendingApproval; SLA clock paused; N-16 sent
MNT-005	PendingApproval → Assigned on approval	Functional	P1	Both	Approval record created	1. Approver approves 2. WO transitions	Status=Assigned; approved_cost written; SLA resumes
MNT-006	PendingApproval → Rejected on denial	Functional	P1	Both	WO awaiting approval	1. Approver rejects with reason	Status=Rejected; WO terminal; requester notified (N-03 analog)
MNT-007	PendingApproval → Diagnosis on re-scope	Functional	P2	Web	WO awaiting approval	1. Approver selects "request re-scope"	Status=Diagnosis; approval cleared; SLA resumes
MNT-008	Block leaving PendingApproval without approval record (V-04)	Negative	P1	API	estimated_cost > tier-0, no approval	1. POST transition to Assigned	422; error "approval required"; status unchanged
MNT-009	Assigned → Repair_Internal (path=INTERNAL)	Functional	P1	Both	Technician + INTERNAL path assigned	1. Assign tech 2. Start repair	Status=Repair_Internal; checklist template applied
MNT-010	Assigned → Vendor_Service (path=VENDOR)	Functional	P1	Both	Vendor assigned, VENDOR path	1. Dispatch vendor 2. Transition	Status=Vendor_Service; SLA paused (vendor); N-19 sent
MNT-011	Assigned → Warranty_Claim (path=WARRANTY)	Functional	P1	Web	In-force warranty, WARRANTY path	1. Select warranty path 2. Transition	Status=Warranty_Claim; SLA paused
MNT-012	Cannot leave Assigned without technician (schema req)	Negative	P2	API	No assigned_technician_id	1. POST transition from Assigned	422; "technician required"; status unchanged
MNT-013	Repair_Internal → AwaitingParts and back	Functional	P2	Both	Repair in progress, parts needed	1. Mark parts needed 2. Receive parts 3. Resume	AwaitingParts pauses SLA; return to Repair_Internal resumes SLA
MNT-014	Testing → QualityCheck on pass	Functional	P1	Both	Repair done, tests recorded	1. Record functional test pass 2. Transition	Status=QualityCheck
MNT-015	Testing → Diagnosis on fail (rework loop)	Functional	P2	Both	Test result = fail	1. Record test fail 2. Transition	Status=Diagnosis; rework_count incremented; N-22 sent
MNT-016	QualityCheck → Complete (QC pass + e-sign)	E2E	P1	Both	Valid e-signature exists, QC pass	1. QC sign-off 2. Complete	Status=Complete; actual_cost rolled up; downtime clock stopped
MNT-017	QualityCheck → Repair_Internal on QC fail	Functional	P2	Web	QC checklist fails	1. QC marks fail	Status=Repair_Internal; rework_count incremented
MNT-018	Complete → BackToInventory (V-17)	Functional	P1	Both	WO Complete	1. Return-to-inventory	Asset status=Available; out-of-service flag cleared; downtime_end set
MNT-019	Return-to-inventory blocked from non-Complete (V-17)	Negative	P2	API	WO in Testing	1. POST return-to-inventory	422; "only from COMPLETE"; asset unchanged
MNT-020	Illegal transition rejected (guarded state machine)	Negative	P1	API	WO in Reported	1. POST transition to Complete	409/422; illegal transition rejected; audit records attempt
MNT-021	Cannot reach Testing with unresolved Fail checklist item (V-07)	Negative	P1	Both	Checklist has Fail item without note/action	1. Attempt transition to Testing	422; blocked until note + corrective action supplied
MNT-022	Cannot reach QualityCheck without e-signature (V-08)	Negative	P1	API	No technician e-signature	1. POST transition to QualityCheck	422; "e-signature required"; status unchanged
MNT-023	Cannot Complete with open vendor ticket (V-10)	Negative	P1	Both	Vendor ticket status != Closed	1. Attempt Complete	422; blocked until vendor ticket Closed
MNT-024	P1 priority auto-sets Critical severity + immediate triage SLA (V-16)	Functional	P2	API	New WO priority=P1	1. Create WO with P1	severity=Critical; response SLA set to immediate policy

6.2 Cost Threshold, SLA Timers & Escalation (MNT)

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
MNT-025	SLA pauses in PendingApproval and resumes on exit (V-12)	Functional	P1	API	WO enters PendingApproval at T	1. Record remaining SLA 2. Approve after delay 3. Compare	Elapsed during pause not counted; sla_resolution_due shifted by pause duration
MNT-026	SLA pauses in AwaitingParts / VendorService / WarrantyClaim (V-12)	Functional	P1	API	WO cycles through each paused state	1. Enter each state 2. Exit 3. Verify clock	Clock frozen in each; resumes on exit; only active-state time accrues
MNT-027	SLA escalation at 50% notifies assignee	Functional	P2	Both	Active resolution SLA reaches 50%	1. Advance clock to 50%	N-20 to assignee (in-app/email/push); no manager alert yet
MNT-028	SLA escalation at 80% notifies manager	Functional	P2	Both	SLA reaches 80%	1. Advance clock to 80%	N-20 escalates to manager

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
MNT-029	SLA 100% sets breach flag + escalation (N-21)	Functional	P1	Both	SLA reaches 100%	1. Advance clock to 100%	sla_breached=true; N-21 to manager+assignee; SMS on P1; webhook fired
MNT-030	Boundary: estimated_cost exactly at tier-0 threshold	Boundary	P2	API	estimated_cost == tier-0 threshold	1. Submit diagnosis	Auto-approve applies ($\leq$ threshold); no PendingApproval
MNT-031	Negative estimated/approved/actual cost rejected (V-15)	Negative	P2	API	estimated_cost = -5	1. POST diagnosis	422; "cost $\geq$ 0"; rejected
MNT-032	actual_cost reconciliation within tolerance (V-13)	Functional	P2	API	labor+parts+vendor sum known	1. Complete WO 2. Read actual_cost	Equals Σ (labor+parts+vendor) within $\pm$ configured %; else flagged
MNT-033	Currency mismatch vs tenant/contract rejected (V-15)	Negative	P3	API	WO currency != tenant currency	1. Submit cost in wrong currency	422; currency validation error

6.3 Parts Consumption & Inventory (MNT)

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
MNT-034	Reserve → consume decrements Inventory (V-05)	Functional	P1	Both	Spare stock available	1. Reserve qty 2. Consume 3. Check stock	Atomic stock decrement = consumed qty on commit
MNT-035	Reserve above available stock blocked (V-05)	Negative	P1	API	quantity_reserved > available	1. POST parts reserve exceeding stock	422; " $\leq$ available inventory"; no reservation
MNT-036	quantity_consumed > quantity_reserved blocked (V-05)	Negative	P1	API	Reserved=2	1. Consume 3	422; "consumed $\leq$ reserved"; stock unchanged
MNT-037	Serialized part requires serials matching qty (V-06)	Negative	P1	Both	Serialized SKU, consume qty=2	1. Consume with 1 serial	422; serial count must equal quantity_consumed
MNT-038	Consumption crossing reorder point fires low-stock (N-29)	Functional	P2	Both	Stock near reorder point	1. Consume qty crossing reorder	Inventory low-stock event; N-29 to Store/Procurement; webhook
MNT-039	Release reservation returns stock	Functional	P2	API	Reservation exists	1. DELETE reservation	Reserved qty released back to available; no decrement
MNT-040	Real-time parts availability lookup before reserve	Functional	P3	Mobile	Technician on WO	1. Scan/open part 2. View availability	Live Inventory availability shown before reserve action

6.4 E-Signature, QC & Warranty/Vendor Guards (MNT)

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
MNT-041	E-signature required to reach Complete + tamper hash (V-08)	Functional	P1	Both	Technician completes work	1. Capture e-sign 2. Verify record	esignature_id set; signed_hash = SHA-256 of WO snapshot; ip + timestamp stored
MNT-042	Tamper hash mismatch detected on frozen e-sign	Audit	P1	API	Signed WO snapshot exists	1. Alter WO field post-sign 2. Recompute hash	Hash mismatch detected/flagged; e-sign frozen and immutable
MNT-043	Independent QC enforced when qc_independent=true (V-09)	RBAC	P1	Web	Tenant policy qc_independent=true; QC actor = assigned tech	1. Same technician attempts QC	403; segregation-of-duties block; QC requires different user
MNT-044	Independent QC allowed by different user	Functional	P2	Web	qc_independent=true; QC by other tech/manager	1. Different actor performs QC	QC accepted; independence flag recorded
MNT-045	Warranty claim requires in-force warranty (V-03)	Negative	P1	Both	warranty_end < today	1. Attempt WARRANTY path	422; "warranty lapsed"; path blocked
MNT-046	Warranty claim succeeds with valid coverage (V-03)	Functional	P2	Web	warranty_end $\geq$ today	1. Submit warranty claim	Claim submitted; RMA tracked; status=Warranty_Claim
MNT-047	AMC type requires in-force contract (V-02)	Negative	P2	API	No/expired amc_contract_id	1. Create WO type=AMC	422; requires valid in-force AMC covering asset
MNT-048	Vendor-portal actor edits only own WOs (V-14)	RBAC	P1	Both	Vendor A logged in; WO assigned to Vendor B	1. Vendor A opens/edits WO	403; scope filter assigned_vendor_id = actor org; no cross-WO access
MNT-049	Vendor portal action notifies IT liaison (N-36)	Functional	P3	Both	Vendor updates own WO	1. Vendor posts vendor-ticket update	N-36 to IT Manager liaison; A-22 audit entry

6.5 Out-of-Service Cascade & Metrics (MNT)

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
MNT-050	Out-of-service cascade full path	E2E	P1	Both	Asset assigned to user, loaner policy enabled	1. Mark asset out-of-service	Asset=Under Maintenance; user+manager notified (N-24); licenses reclaimed; loaner request raised; downtime clock started; assignment blocked
MNT-051	Assignment blocked while WO open (V-01)	Negative	P1	API	Asset has open WO / Under Maintenance	1. Attempt assign/transfer asset	422; blocked by cascade; asset stays Under Maintenance
MNT-052	No loaner when loaner policy disabled	Functional	P2	Web	Loaner policy disabled, asset assigned	1. Mark out-of-service	No loaner raised; other cascade steps still occur
MNT-053	Reverse cascade on return-to-inventory	E2E	P1	Both	WO Complete, cascade active	1. Return-to-inventory	Asset=Available; downtime clock stops; licenses restored/released; loaner returned
MNT-054	MTTR / uptime computed on valid downtime pair (V-11)	Functional	P2	API	downtime_end $\geq$ downtime_start	1. Complete WO 2. Read reports/mttr, /uptime	MTTR = Σ (end-start)/count; uptime % correct; invalid pairs excluded

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
MNT-055	downtime_end < downtime_start rejected (V-11)	Negative	P3	API	Attempt to set end before start	1. PATCH downtime_end earlier	422; invalid pair excluded from MTTR

6.6 Technician Mobile Execution (MNT — Mobile)

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
MNT-056	My Queue priority sort + SLA countdown	Functional	P2	Mobile	Tech has assigned WOs	1. Open My Queue	WOs priority-sorted; SLA countdown badges accurate
MNT-057	QR scan opens correct WO	Functional	P2	Mobile	Asset tagged, WO exists	1. Scan asset QR	Correct WO opens
MNT-058	Guided checklist pass/fail + measurement	Functional	P2	Mobile	Checklist template applied	1. Step through items 2. Enter measurements 3. Mark Fail with note	Results saved; Fail requires note before proceeding
MNT-059	Photo capture before/during/after attaches to WO	Functional	P3	Mobile	WO open	1. Capture photos each stage	Attachments saved with kind Photo_Before/During/After; ≤25 MB, image mime
MNT-060	Part scan records consumption	Functional	P2	Mobile	Serialized/stocked part	1. Scan part barcode 2. Confirm qty	Consumption recorded; stock decremented; serials captured
MNT-061	On-device e-signature generates hash + PDF	Functional	P1	Mobile	Work complete	1. Sign on pad 2. Submit	signed_hash + ESignPDF generated; frozen to WO
MNT-062	Offline mode caches WOs, queues actions, syncs	E2E	P1	Mobile	Tech offline with cached WOs	1. Complete checklist offline 2. Reconnect	Actions queued; sync on reconnect; conflicts resolved
MNT-063	Offline conflict resolution on reconnect	Negative	P2	Mobile	Same WO edited server-side during offline window	1. Reconnect with local changes	Conflict detected; resolution applied per policy; no silent overwrite

6.7 Approvals — Chain, Decisioning, SoD, Delegation (APR)

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
APR-001	Multi-step chain materialized at submit	Functional	P1	Both	Tenant chain: Manager→Finance	1. Submit approval-triggering WO/request	Full step chain instantiated at submit time; steps ordered per config
APR-002	Threshold inserts Finance step (BR-05)	Functional	P1	Both	Cost ≥ Finance threshold	1. Submit request above threshold	Mandatory Finance step inserted after managerial approval
APR-003	Below threshold: no Finance step	Boundary	P2	API	Cost just below threshold	1. Submit	Only managerial step present; no Finance insertion
APR-004	Boundary: cost exactly at threshold triggers Finance (BR-05)	Boundary	P2	API	Cost == threshold	1. Submit	Finance step inserted (≥ threshold inclusive)
APR-005	Approver-type USER decisioning	Functional	P2	Both	Step approver-type=USER	1. Named user approves	Step resolved by that specific user only
APR-006	Approver-type LINE_MANAGER resolves to requester's manager	Functional	P2	Both	Step type=LINE_MANAGER	1. Requester's manager approves	Correct manager resolved; others cannot approve step
APR-007	Approver-type ROLE — any role holder can decide	Functional	P2	Both	Step type=ROLE	1. Any user with role approves	Step resolved by any qualifying role member
APR-008	Approver-type DEPARTMENT_HEAD resolution	Functional	P3	Both	Step type=DEPARTMENT_HEAD	1. Dept head approves	Resolved to requester's department head
APR-009	SoD: approver ≠ requester (BR-04)	RBAC	P1	Both	Requester also holds approver role	1. Requester attempts to approve own request	403; SoD block; step stays pending
APR-010	Delegation preserves SoD — delegate cannot approve own (BR-06)	RBAC	P1	Both	Delegate is also the requester	1. Delegate opens own request approval	403; delegate blocked on own request
APR-011	Valid delegation lets delegate approve others	Functional	P2	Both	Delegation configured, request from third party	1. Delegate approves	Approval accepted; recorded as delegated decision
APR-012	Approval SLA escalation (BR-06)	Functional	P2	Both	Pending approval past SLA	1. Advance clock past approval SLA	Escalation fired to next approver/manager; audit logged
APR-013	One-tap approve on mobile push	Functional	P1	Mobile	Pending approval, push received	1. Tap approve in push/app	Decision recorded; chain advances; other channels reconcile
APR-014	Web inbox approve mirrors mobile	Functional	P2	Web	Pending approval in web inbox	1. Approve from inbox	Same decision path; idempotent with mobile
APR-015	Reject halts chain and notifies requester	Functional	P2	Both	Multi-step chain in progress	1. Any step rejects	Chain stops; downstream steps not materialized/executed; requester notified
APR-016	Cross-tenant approval isolation (BR-03)	RBAC	P1	API	Approver in tenant A, request in tenant B	1. Attempt to load/approve	403/404; tenant filter blocks; no cross-tenant visibility

6.8 Notifications (NTF)

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
NTF-001	WO created recipients/channels (N-15)	Functional	P2	Both	New WO created	1. Create WO	Assigned tech + manager get in-app+email+push; webhook fired
NTF-002	Maintenance approval request (N-16) mandatory	Functional	P1	Both	Cost-threshold WO enters PendingApproval	1. Trigger approval	N-16 to Manager/Office Admin via in-app+email+push; cannot be disabled
NTF-003	Assigned/reassigned push only (N-17)	Functional	P3	Mobile	Tech assigned	1. Assign WO	In-app + push to technician; no email per matrix

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
NTF-004	Awaiting parts notifies tech + store (N-18)	Functional	P3	Both	WO enters AwaitingParts	1. Transition	N-18 in-app+email+webhook to technician and store
NTF-005	SLA at-risk 50/80 recipients differ (N-20)	Functional	P2	Both	SLA thresholds crossed	1. Reach 50% then 80%	50% → assignee; 80% → manager; mandatory, cannot disable
NTF-006	SLA breached SMS on P1 (N-21)	Functional	P1	Both	P1 WO breaches SLA	1. Reach 100%	In-app+email+push+SMS(P1)+webhook to manager+assignee; mandatory
NTF-007	Asset out-of-service to affected user (N-24) mandatory	Functional	P1	Both	Asset OOS with assigned user	1. Mark OOS	N-24 to affected user + manager; includes loaner status; cannot be disabled
NTF-008	Mandatory notification cannot be disabled by preference	Negative	P1	Both	User disables N-16/N-20/N-21 in prefs	1. Attempt to opt out 2. Trigger event	Preference toggle blocked/ignored for mandatory events; still delivered
NTF-009	Non-mandatory notification respects opt-out	Functional	P3	Both	User disables N-17 (assigned)	1. Opt out 2. Assign WO	Optional alert suppressed per preference
NTF-010	Digest coalescing — seat limit (N-11)	Functional	P2	Both	Multiple seat-limit events in window	1. Trigger repeated N-11 events	Coalesced into digest, not per-event spam; correct recipients
NTF-011	Digest coalescing — SLA breach batch	Functional	P2	Both	Many SLA breaches in interval	1. Trigger burst	Batched digest to manager; individual mandatory alerts still honored
NTF-012	Digest coalescing — offboarding overdue (N-32)	Functional	P2	Both	Return overdue daily cadence	1. Advance to Due+1, next day	Daily coalesced overdue digest; SMS+push+email+webhook; mandatory
NTF-013	Preventive due lead-time notification (N-25)	Functional	P3	Both	PM scheduled, lead-time reached	1. Reach lead-time (e.g. 7-day)	N-25 to assigned tech + manager; webhook
NTF-014	Predictive health alert (N-26)	Functional	P3	Both	Health score below threshold	1. Ingest degraded signal	N-26 to IT Manager + technician; in-app+email+push+webhook
NTF-015	Webhook payload for wo.sla_breached	Functional	P3	API	Tenant webhook configured	1. Breach SLA	maintenance.wo.sla_breached delivered to Teams/Slack/ITSM endpoint

6.9 Audit (AUD)

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
AUD-001	WO state transition append-only (A-14)	Audit	P1	API	WO transitions state	1. Transition 2. Read history	audit_event with from_state→to_state, actor context; no update/delete
AUD-002	Maintenance decision before/after diff (A-13)	Audit	P1	API	Approve/reject/re-scope	1. Decide 2. Inspect event	est→approved cost, decision, reason captured; 5-yr retention
AUD-003	Hash-chain tamper evidence (prev_hash/event_hash)	Audit	P1	API	Multiple events per tenant	1. Read chain 2. Alter one event	prev_hash+event_hash link intact; tampering breaks chain and is detectable
AUD-004	E-signature frozen/immutable (A-16)	Audit	P1	API	e-sign captured	1. Attempt to edit/delete e-sign event	Rejected; record frozen; 7-yr retention
AUD-005	Parts consumption stock before→after (A-15)	Audit	P2	API	Parts consumed	1. Consume 2. Read audit	qty, serials, stock before→after captured
AUD-006	QC sign-off with independence flag (A-17)	Audit	P2	API	QC performed	1. Sign off 2. Read event	result, signer, independence flag recorded
AUD-007	Vendor-portal action audited (A-22)	Audit	P2	API	Vendor edits own WO	1. Vendor updates ticket	field diff + external ref + cost logged; 3rd-party audit tag
AUD-008	Config change (SLA/threshold) audited (A-18)	Audit	P2	Both	Admin edits SLA threshold	1. Change config	before→after setting captured; N-34 sent; 3-yr retention
AUD-009	Data export logged (A-20)	Audit	P2	Both	Authorized export of WO report	1. Export report	entity, filter, row count, format logged; N-35 to Admin/DPO
AUD-010	Auditor read-only writes only to findings store	RBAC	P1	Both	Auditor role, time-boxed scope	1. Attempt to edit WO/audit ledger 2. Add a finding	WO/ledger edits denied; findings write allowed to separate findings store only
AUD-011	Append-only ledger — no update/delete	Negative	P1	API	Existing audit_event	1. Attempt UPDATE/DELETE on ledger	Rejected; corrections only via new compensating event
AUD-012	Retention honored per action class	Audit	P3	API	Events of varying classes	1. Run retention job	A-16/e-sign 7-yr, A-13/14 5-yr, A-18 3-yr enforced; deletion log itself retained (A-24)
AUD-013	Actor context completeness	Audit	P3	API	Any state-changing write	1. Read event	actor_id, actor_role, on_behalf_of, source_ip, session_id, tenant_id, UTC timestamp, request_id present

7. Mobile, Offline Sync & Web↔Mobile Parity — Test Scenarios

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
MOB-001	Offline read of scoped assets/locations	Functional	P1	Mobile	Device synced online once; scoped assets cached in SQLite; airplane mode on	1. Enable airplane mode 2. Open Assets tab 3. Search tag "AST-1021" 4. Open detail	Asset list, detail, photos, history render fully from SQLite with no network; offline banner + sync badge visible
MOB-002	Read of out-of-scope asset offline	Negative	P2	Mobile	Offline; asset NOT in device scope	1. Airplane mode 2. Search a tag outside scoped locations	Graceful empty/"not available offline" state; no crash; option to retry when online
MOB-003	Queue assign action offline	Functional	P1	Mobile	Offline; asset In Service	1. Scan asset 2. Assign → pick employee 3. Capture condition + photo + e-sign 4. Confirm	Op enqueued in SQLite with opId, baseVersion, delta; row shows "pending sync"; badge count increments; optimistic UI shows Assigned
MOB-004	Queue return + condition photo offline	Functional	P1	Mobile	Offline; asset Assigned	1. Scan 2. Return → condition Good 3. Add photo 4. Confirm	Return op + photo queued separately; asset optimistically In Service; sync badge reflects both ops
MOB-005	Queue offline stock count session	Functional	P1	Mobile	Offline; count session started for location	1. Start count 2. Scan-to-increment 15 items 3. View variance 4. Submit	Session + increments stored locally; running total correct; submit queues additive count op; variance computed vs cached expected
MOB-006	Queue e-signature acknowledgment offline	Functional	P1	Mobile	Offline; assign flow at sign step	1. Draw signature 2. Confirm	Signed ack captured with name, timestamp, GPS, asset snapshot; queued; syncs as signed acknowledgment on reconnect
SYNC-001	Idempotent replay — no duplicate on retry	Functional	P1	Both	3 ops queued offline; flaky network causing one retry	1. Reconnect 2. Sync batch 3. Force retry of a timed-out op (same opId)	Server dedups by Idempotency-Key=opId; first result returned on replay; exactly one state change; no duplicate assignment/history rows
SYNC-002	Optimistic version conflict → 409 + server state	Negative	P1	Both	Asset edited on web (version bumped) after mobile captured baseVersion	1. Mobile replays op with stale baseVersion 2. Server compares	409 conflict returned with current server state and version; op marked "conflict" in Sync Center, not silently dropped
SYNC-003	Field-level LWW resolution	Functional	P1	Both	Conflict on non-overlapping fields (mobile changed condition, web changed location)	1. Trigger sync conflict 2. Auto-resolve	Field-level last-writer-wins merges both changes; state-transition fields defer to server authority; audit records merge
SYNC-004	User-prompt on irreconcilable field clash	Usability	P1	Mobile	Same field changed on both web and mobile	1. Sync conflicting op 2. Conflict prompt appears in Sync Center	User shown both values (mine vs server) with timestamps; choosing a resolution commits it; decision logged in audit
SYNC-005	Dependent-op ordering create→assign	Functional	P1	Both	Offline: new asset created then immediately assigned	1. Create asset offline 2. Assign it offline 3. Reconnect + sync	Server applies create before assign (ordered by entity + client-sequence); assign resolves the newly created server id; no orphan/FK error
SYNC-006	Ordering preserved across mixed entities	Boundary	P2	API	Queue with interleaved create/assign/return/count ops	1. Reconnect 2. Batch sync	Ops applied per-entity in client-sequence order; cross-entity dependencies honored; per-op results returned
MOB-007	Attachment resumable upload interrupted	Functional	P1	Mobile	Large condition photo queued; connection drops mid-upload	1. Start sync 2. Kill network at 50% 3. Restore network	Upload resumes from last chunk (not restart); record link reconciled after completion; no duplicate blob
MOB-008	Attachment upload — record link reconciliation	Functional	P2	Both	Photo uploaded after its parent op already synced	1. Sync op 2. Later complete photo upload	Object-storage link attached to correct asset/history record after upload; appears on web detail
SYNC-007	Delta pull since cursor	Functional	P1	Both	Device has cursor from last sync; web made changes since	1. Call /sync/changes?since=<cursor>	Only records changed after cursor for device-scoped assets/locations returned; new cursor issued; no full re-download
SYNC-008	Delta pull with empty cursor (first sync)	Boundary	P2	API	Fresh device, no cursor	1. Request changes with no/base cursor	Full scoped snapshot returned within paging limits; stable cursor established
MOB-009	QR code scan → asset lookup	Functional	P1	Mobile	Asset tagged with QR; camera permission granted	1. Tap Scan FAB 2. Align QR	Haptic+audio confirm; bottom sheet shows asset summary + contextual actions within 1s; works offline from cache
MOB-010	Code-128 barcode scan	Functional	P2	Mobile	Item labeled Code-128	1. Scan barcode	Decoded correctly; resolves to asset/SKU; unsupported/garbled code shows retry, no crash
MOB-011	Batch scan mode	Functional	P2	Mobile	Batch toggle on; multiple items	1. Enable batch 2. Scan 10 items rapidly	Each hit added to running list with dedupe on repeat scan; count accurate; torch + haptic per hit
MOB-012	Scan invalid/unknown code	Negative	P2	Mobile	Code not in system	1. Scan unknown QR	Clear "not found" message with option to create/link; no silent failure
MOB-013	Biometric unlock	Functional	P1	Mobile	Biometric enrolled; device key in enclave	1. Launch app 2. Face/Touch prompt	Unlock via device-scoped key; PIN fallback offered on biometric fail; session established
MOB-014	Step-up biometric for sensitive op (disposal)	Security	P1	Mobile	Logged in; attempt disposal/decommission	1. Trigger disposal 2. Step-up prompt	Re-auth (biometric) required before commit; cancel aborts op; step-up recorded in audit
MOB-015	Biometric revoked / device remote-wiped	Negative	P2	Both	Admin revokes device from web	1. Admin revokes 2. Mobile attempts action	Device session invalidated; app forces re-auth/logout; queued ops blocked until re-registration
MOB-016	Push deep-link to screen	Functional	P1	Mobile	Approval push received	1. Tap notification	Deep-links directly to the relevant request/asset screen (not just home); context preloaded
MOB-017	Inline approve/reject from push	Functional	P1	Mobile	Actionable approval push	1. Long-press/expand notification 2. Tap Approve	Decision submitted (idempotent); optimistic UI + queued if offline; requester notified on commit
SYNC-009	Mobile change appears on web after sync (assignment)	E2E	P1	Both	Mobile assigns asset offline then reconnects	1. Assign on mobile 2. Sync 3. Refresh web asset list	Web shows asset Assigned to chosen employee, correct timestamp, history row, and signature link

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
SYNC-010	Web change appears on mobile after delta pull (request status)	E2E	P1	Both	Approver approves request on web	1. Approve on web 2. Mobile pulls delta	Mobile request list shows Approved status; push received; no stale Pending state
SYNC-011	Stock count on mobile reflects on web review	E2E	P1	Both	Offline count submitted from mobile	1. Submit count on mobile 2. Sync 3. Open web Count Session Detail	Web shows counted quantities, variance vs expected, and reconciled totals matching mobile
SYNC-012	Approval decision two-way parity	E2E	P1	Both	Same request visible on web and mobile	1. Approve on mobile 2. Sync 3. Check web AND re-open mobile from another device	Decision consistent everywhere; double-approval prevented (idempotent); audit shows single decision
SYNC-013	Concurrent edit both surfaces same asset	Negative	P2	Both	Web and mobile edit same asset field simultaneously	1. Edit on both 2. Sync mobile	Loser gets 409 + server state; resolution flow engaged; no lost-update silently overwriting
MOB-018	Sync Center manual "sync now" + status states	Usability	P2	Mobile	Ops in queued/syncing/conflict states	1. Open Sync Center 2. Tap Sync Now	Each op shows correct status (queued/syncing/conflict/done); manual trigger flushes queue; conflicts remain actionable

8. UI/UX Validation Checklist — Web AND Mobile

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
UI-001	Login — light & dark mode rendering	Usability	P1	Both	App at login screen	1. Load login 2. Toggle system light→dark	Both themes render with correct tokens; no unstyled flash; logo/inputs legible in both
UI-002	Login — form validation messaging	Usability	P1	Both	Login form	1. Submit empty 2. Enter bad credentials	Inline field-level errors (not a disabled button with no reason); error summary announced; submit stays enabled to re-try
UI-003	Dashboard — loading/empty/error/success states	Usability	P1	Both	Role dashboard	1. Load with slow net 2. Load tenant with no data 3. Force API error	Skeleton loaders while loading; friendly empty state with CTA; error state with retry; populated success state — all four exist
UI-004	Dashboard — KPI tile contrast WCAG AA	Accessibility	P1	Both	Dashboard KPI strip	1. Measure text/bg contrast light+dark	All KPI text ≥ 4.5:1 (≥3:1 large); delta colors not sole signal (arrow/label present)
UI-005	Asset list — responsive breakpoint grid→cards	Usability	P1	Web	Asset list with data	1. Resize desktop→tablet→<768px	Data grid reflows to cards below 768px; nav collapses to hamburger <1024px; no horizontal body scroll
UI-006	Asset list — RN phone size adaptation	Usability	P1	Mobile	Asset list on small + large phone	1. View on 375px and 430px widths	Layout adapts; no clipped text; safe-area insets respected (notch/home indicator)
UI-007	Asset list — status badges legible & not color-alone	Accessibility	P1	Both	Assets across statuses	1. Inspect badges 2. Simulate color-blindness	Each badge has dot + text label (icon+label); distinguishable in grayscale; AA contrast
UI-008	Asset detail — keyboard nav + focus rings	Accessibility	P1	Web	Asset detail drawer/page	1. Tab through tabs, fields, actions	Logical focus order; visible focus ring on every interactive element; ESC closes drawer; focus returns to trigger
UI-009	Asset detail — screen-reader labels	Accessibility	P1	Both	SR (NVDA/VoiceOver) on detail	1. Navigate with SR	Tabs, status chip, action buttons have descriptive accessible names; images have alt/label; timeline readable
UI-010	Asset detail — 44px touch targets	Accessibility	P1	Mobile	Detail action bar	1. Measure tap targets	Assign/Return/Transfer/Report buttons ≥ 44×44px with adequate spacing; no mis-tap adjacency
UI-011	License — over-deploy error state clarity	Usability	P1	Both	AutoCAD over-deployed by 3	1. Attempt assign past limit	Blocking error card with icon+label, plain-language reason, and actionable options (Buy/Reclaim/Cancel); not just a greyed button
UI-012	License — seat-pool meter accessibility	Accessibility	P2	Both	Seat-pool cards	1. Inspect used/free meters	Progress conveyed by number+label, not color/fill alone; at-risk/over states iconized; AA contrast
UI-013	Procurement — PO kanban keyboard/drag alt	Accessibility	P1	Web	PO board	1. Move card without mouse	Keyboard-accessible move (menu or arrow) alternative to drag; SR announces column + status change
UI-014	Procurement — loading/empty/error on board	Usability	P2	Both	PO board	1. Load slow 2. No POs 3. API error	Column skeletons; empty column messaging; error with retry — all present
UI-015	Inventory — form validation inline messaging	Usability	P1	Both	Adjust/receive stock form	1. Enter negative/invalid qty 2. Submit	Inline validation with clear message near field; submit not silently disabled; valid state clears error
UI-016	Maintenance — status/priority badges not color-alone	Accessibility	P1	Both	Ticket with P1/In Progress	1. Inspect priority + status chips	Priority and SLA state carry text/icon; grayscale-distinguishable; SLA meter has label
UI-017	Approvals — success/confirmation state	Usability	P2	Both	Approval inbox	1. Approve a request	Clear success toast/state; row updates; undo or confirmation as designed; no ambiguous silent result
UI-018	Command palette (⌘K) operation	Usability	P2	Web	Any web screen	1. Press ⌘K 2. Search asset/action/screen	Palette opens focus-trapped; keyboard-only searchable; Enter executes; ESC closes; SR announces
UI-019	Pull-to-refresh on lists	Usability	P2	Mobile	Assets/Inventory/Tasks list	1. Pull down at top	Refresh spinner; data re-fetches/delta-pulls; haptic; works and settles without layout jump
UI-020	Mobile scan/assign sheet — states + safe area	Usability	P1	Mobile	Assign bottom sheet	1. Open sheet 2. Trigger loading/error on submit	Sheet respects safe-area; shows submitting spinner, error retry, success dismiss; 44px controls; swipe-down dismiss
UI-021	Reduced-motion preference respected	Accessibility	P2	Both	OS reduce-motion enabled	1. Trigger drawer/sheet/skeleton animations	Animations minimized/removed; no parallax or large motion; functionality intact
UI-022	Dark-mode parity across all key screens	Usability	P2	Both	Every listed screen	1. Toggle dark on each screen	Full theme parity; no hardcoded light-only colors; charts use theme-aware series palette; contrast holds
UI-023	Charts — empty/loading/error + colorblind palette	Accessibility	P2	Both	Dashboard charts	1. Load, empty, error each chart	Each has loading/empty/error state; colorblind-safe series with data labels + legend; not color-only
UI-024	Forms — error not conveyed by color alone	Accessibility	P1	Both	Any validated form	1. Trigger validation error	Error shown with icon + text + border, aria-invalid + aria-describedby; SR announces the message

9. Non-Functional & Cross-Cutting — Test Scenarios

ID	Scenario	Type	Pri	Surface	Preconditions	Steps	Expected Result
NFR-001	p95 interactive read latency	Performance	P1	API	Load-test rig; representative tenant	1. Drive sustained read load on asset/list endpoints 2. Measure	p95 < 300 ms, p99 < 800 ms for interactive reads under target concurrency
NFR-002	List pagination at scale	Performance	P1	API	Tenant with 2M+ assets	1. Cursor-page through list with filters 2. Measure per-page latency	Stable cursor paging; consistent latency across deep pages; no offset degradation; no duplicate/missing rows
NFR-003	Discovery burst ingest	Performance	P1	API	Discovery agent feeding 100k+ records	1. Send burst 2. Monitor queue depth + workers	Ingest absorbed via BullMQ without API latency breach; workers scale on queue depth; no data loss; reconciliation completes
NFR-004	Security headers / CSP present	Security	P1	Web	App served	1. Inspect response headers	CSP, X-Content-Type-Options, Referrer-Policy, Permissions-Policy, HSTS present; per-tenant CORS allow-list enforced
NFR-005	Upload magic-byte + AV scan	Security	P1	Both	Upload endpoint	1. Upload disguised executable (wrong extension) 2. Upload EICAR test file	File-type validated by magic bytes not extension; AV scan quarantines EICAR; malicious upload rejected with problem+json
NFR-006	Rate-limit + brute-force logout	Security	P1	API	Login + API endpoints	1. Exceed sliding-window on route-class 2. Repeated bad logins	429 with RateLimit-/Retry-After; account/IP logout after threshold; interactive vs bulk vs agent buckets separate
NFR-007	No PII in URLs / query strings	Security	P1	Both	Instrumented traffic capture	1. Exercise flows with personal data	No SSN/email/serial/PII in URLs, query params, or logs; sensitive lookups use body/headers
NFR-008	Secret handling — none in images/env/logs	Security	P1	API	Built container + runtime logs	1. Scan image + env + logs for secrets	Secrets sourced from Vault/KMS only; no plaintext secrets in images, env files, or logs; per-tenant creds envelope-encrypted
NFR-009	Idempotency-Key replay protection	Security	P2	API	Mutation endpoint	1. Replay identical POST with same Idempotency-Key within 24h	First result returned; no duplicate side effect; distinct key creates new resource
NFR-010	Backup + tested restore (RPO/RTO)	Reliability	P1	API	Backup + PITR configured; staging restore target	1. Take backup 2. Simulate loss 3. Restore via PITR 4. Measure	Data restored to ≤ 5 min RPO; full restore within ≤ 1 h RTO; integrity verified post-restore
NFR-011	DR failover to cross-region replica	Reliability	P2	API	Cross-region replica live	1. Fail primary region 2. Promote replica	Service resumes within RTO; minimal data loss within RPO; residency constraints preserved
NFR-012	Tenant isolation under load	Security	P1	API	Multiple tenants; concurrent load	1. Drive queries across tenants with RLS 2. Attempt cross-tenant id access	RLS blocks all cross-tenant access even under load/buggy query; noisy-neighbor rate-limits prevent starvation; zero data bleed
NFR-013	Tenant cache/queue isolation	Security	P2	API	Redis cache + queues shared	1. Exercise cached reads + queued jobs for 2 tenants	Cache keys tenant-prefixed (t:(tenantId)); per-tenant queue rate groups; no cross-tenant cache hit or job leakage
NFR-014	Accessibility audit gate (WCAG 2.1 AA)	Accessibility	P1	Both	CI a11y audit configured	1. Run automated + manual audit on key screens	No critical/serious WCAG AA violations; gate blocks merge on regressions; manual keyboard/SR spot-checks pass
NFR-015	i18n / localization	Usability	P2	Both	Locale switch + RTL locale	1. Switch language 2. Check dates/numbers/currency 3. Check RTL	Strings externalized/translated; locale-correct date/number/currency; layout holds under longer strings and RTL mirroring
NFR-016	Observability — trace correlation	Functional	P2	API	OpenTelemetry enabled	1. Issue request that spans API→queue→worker→DB	Single tracelid correlates request→queue→worker→DB spans; tracelid surfaced in problem+json errors; per-tenant dashboards populate
NFR-017	Availability SLO under sustained load	Performance	P2	API	Steady-state load test	1. Run extended soak 2. Measure error rate/uptime	Meets 99.9% (enterprise 99.95%) availability; error budget not exhausted; auto-scaling holds SLO
NFR-018	Zero-downtime migration	Reliability	P2	API	Expand/contract migration staged	1. Deploy migration during live traffic	No request errors during migration; reversible; reads/writes continue; feature flag gates rollout
NFR-019	Webhook HMAC signature verification	Security	P2	API	Tenant webhook subscription	1. Deliver event 2. Tamper payload	Valid HMAC-SHA256 signature with timestamp accepted; tampered/expired signature rejected; retries + DLQ + replay work
NFR-020	Step-up MFA on sensitive bulk ops	Security	P1	Web	Bulk delete / cost export	1. Trigger bulk delete or cost export	Step-up MFA (TOTP/WebAuthn/push) required before execution; denial blocks op; action + step-up audited

I've written the three test scenario sections. Summary of what's delivered:

- **Section 7 (Mobile, Offline Sync & Web↔Mobile Parity):** 30 scenarios (**MOB-###** / **SYNC-###**) covering offline scoped reads, offline action queuing (assign/return/count/photo/e-sign), idempotent replay with no-duplicate-on-retry, 409 version-conflict resolution (field-level LWW + user-prompt), dependent-op ordering (create→assign), resumable attachment upload, delta pull by cursor, QR/Code-128/batch scanning, biometric unlock + step-up, push deep-links + inline approve, and explicit two-way parity checks (mobile→web and web→mobile for assignment, request status, stock count, approval decision).
- **Section 8 (UI/UX Validation Checklist):** 24 rows (**UI-###**), mostly Surface=Both or paired Web/Mobile, covering light/dark, responsive/adaptive layout, loading/empty/error/success states, inline validation (not silently-disabled buttons), status badges not color-alone, keyboard nav + focus rings, 44px targets + safe-area, WCAG AA contrast, SR labels, reduced-motion, command palette, and pull-to-refresh across login, dashboard, asset list/detail, license, procurement, inventory, maintenance, approvals, and the mobile scan/assign sheet.
- **Section 9 (Non-Functional & Cross-Cutting):** 20 scenarios (**NFR-###**) covering p95/p99 latency, pagination at scale, discovery burst, CSP/security headers, upload magic-byte/AV, rate-limit/lockout, no-PII-in-URLs, secret handling, backup/tested-restore with RPO≤5m/RTO≤1h, DR failover, tenant isolation under load, cache/queue isolation, a11y audit gate, i18n/RTL, trace correlation, availability SLO, zero-downtime migration, webhook HMAC, and step-up MFA on sensitive ops.

Total: **74 scenarios**. Columns match the exact spec (**ID** | **Scenario** | **Type** | **Pri** | **Surface** | **Preconditions** | **Steps** | **Expected Result**), with the required ID prefixes, Type/Pri/Surface enums, terse numbered steps, and precise pass-condition Expected Results. All scenarios are grounded in the two spec sources (offline op-queue/idempotency/conflict model, NFR targets, UI design-system tokens/components, and the role dashboards/wireframes).

Note: the section count came in slightly above your ~45–55 target because the two-way parity checks and the per-surface UI checklist each needed their own rows; I can trim to a tighter set if you'd prefer closer to 55.